

RESEARCH ARTICLE

Geochemistry and $^{40}\text{Ar}/^{39}\text{Ar}$ geochronology of the Nandurbar-Dhule mafic dyke swarm: Dyke-sill-flow correlations and stratigraphic development across the Deccan flood basalt province

Hetu Sheth¹  | Loïc Vanderkluyzen²  | Elena I. Demonterova³ | Alexei V. Ivanov³ | Valery M. Savatenkov^{4,5}

¹Department of Earth Sciences, Indian Institute of Technology Bombay (IITB), Mumbai, India

²Department of Biodiversity, Earth & Environmental Science, Drexel University, Philadelphia, Pennsylvania, USA

³Institute of the Earth's Crust, Siberian Branch of the Russian Academy of Sciences, Irkutsk, Russia

⁴Department of Geochemistry, Institute of Earth Sciences, St. Petersburg State University, St. Petersburg, Russia

⁵Institute of Precambrian Geology and Geochronology, Russian Academy of Sciences, St. Petersburg, Russia

Correspondence

Hetu Sheth, Department of Earth Sciences, Indian Institute of Technology Bombay (IITB), Powai, Mumbai 400076 India.
Email: hcsheth@iitb.ac.in

Funding information

U.S. National Science Foundation, Grant/Award Numbers: EAR-1250440 and EAR-0229825; Russian Foundation for Basic Research, Grant/Award Number: 15-05-05130; Industrial Research and Consultancy Centre (IRCC), IIT Bombay, Grant/Award Number: 03IR014

Handling Editor: N.V. C. Rao

The ENE-WSW-trending Nandurbar-Dhule swarm is the best developed tholeiitic dyke swarm in the Deccan Traps. We obtained $^{40}\text{Ar}/^{39}\text{Ar}$ ages of 67.06 ± 0.60 , 67.49 ± 0.89 , and 63.43 ± 0.48 Ma (2σ internal errors) on three of its largest dykes (36–79 km long), indicating that swarm emplacement spanned ≥ 2.5 million years under regional crustal extension. Our Sr-Nd-Pb isotopic data, combined with previously available geochemical data, identify multiple magma injections in some dykes and also identify probable feeder dykes of some lavas in the lower Western Ghats sequence and in Saurashtra, each ~ 200 km away. Several dykes are compositionally distinct from hitherto analysed lavas; $>50\%$ of the analysed Nandurbar-Dhule dykes are isotopically like the Mahabaleshwar and Panhala formations of the upper Western Ghats sequence, covering a very narrow isotopic range, but have the distinctive chemical signatures of the high-TiO₂ Kolhapur Unit of the southernmost Western Ghats. These dykes thus possess a unique combination of isotopic and chemical characteristics not hitherto known in Deccan tholeiites, cross-combining features of different eruptive units in the Wai Subgroup of the Western Ghats. This new, “Nandurbar-type” chemical-isotopic flavour is however frequently observed in dykes, sills, and lavas in the Pachmarhi, Shahdol, and Mandla areas 450–600 km to the east, and in Deccan-age dykes cutting through the Early Cretaceous Rajmahal Traps of eastern India. Varied geochemical evidence indicates that the northern and northeastern Deccan lava stratigraphy (such as the Pavagadh section and the Pachmarhi-Shahdol-Jabalpur-Mandla areas) is largely independent of the Western Ghats lava stratigraphy.

KEYWORDS

Deccan Traps, dyke swarms, flood basalts, India, Nandurbar-Dhule

1 | INTRODUCTION

The Late Cretaceous to Palaeocene Deccan flood basalts (~ 66 – 62 Ma, Baksi, 2014; Pande, Yatheesh, & Sheth, 2017; Renne et al., 2015; Schoene et al., 2015) are best exposed in the Western Ghats escarpment near the western Indian rifted margin (Figure 1). Based on field, geochemical, and magnetic polarity criteria, this entirely tholeiitic sequence has been divided into three subgroups and 12 formations with a stratigraphic thickness of ~ 3.4 km over a ~ 500 km distance (e.g., Beane, Turner, Hooper, Subbarao, & Walsh, 1986; Cox & Hawkesworth,

1984, 1985; Devey & Lightfoot, 1986; Khadri, Subbarao, Hooper, & Walsh, 1988; Lightfoot, Hawkesworth, Devey, Rogers, & van Calsteren, 1990; Najafi, Cox, & Sukheswala, 1981; Table 1). Other tholeiitic sequences in the province, such as Toranmal and Ellora-Ajanta-Lonar (see Figure 1 for all localities mentioned in this paper), are broadly correlatable with the Western Ghats type section (e.g., Jay & Widdowson, 2008; Mahoney, Sheth, Chandrasekharam, & Peng, 2000; Peng, Mahoney, Hooper, Macdougall, & Krishnamurthy, 1998; Peng, Mahoney, Hooper, & Vanderkluyzen, 2014; Subbarao, Chandrasekharam, Navaneethakrishnan, & Hooper, 1994). Yet other

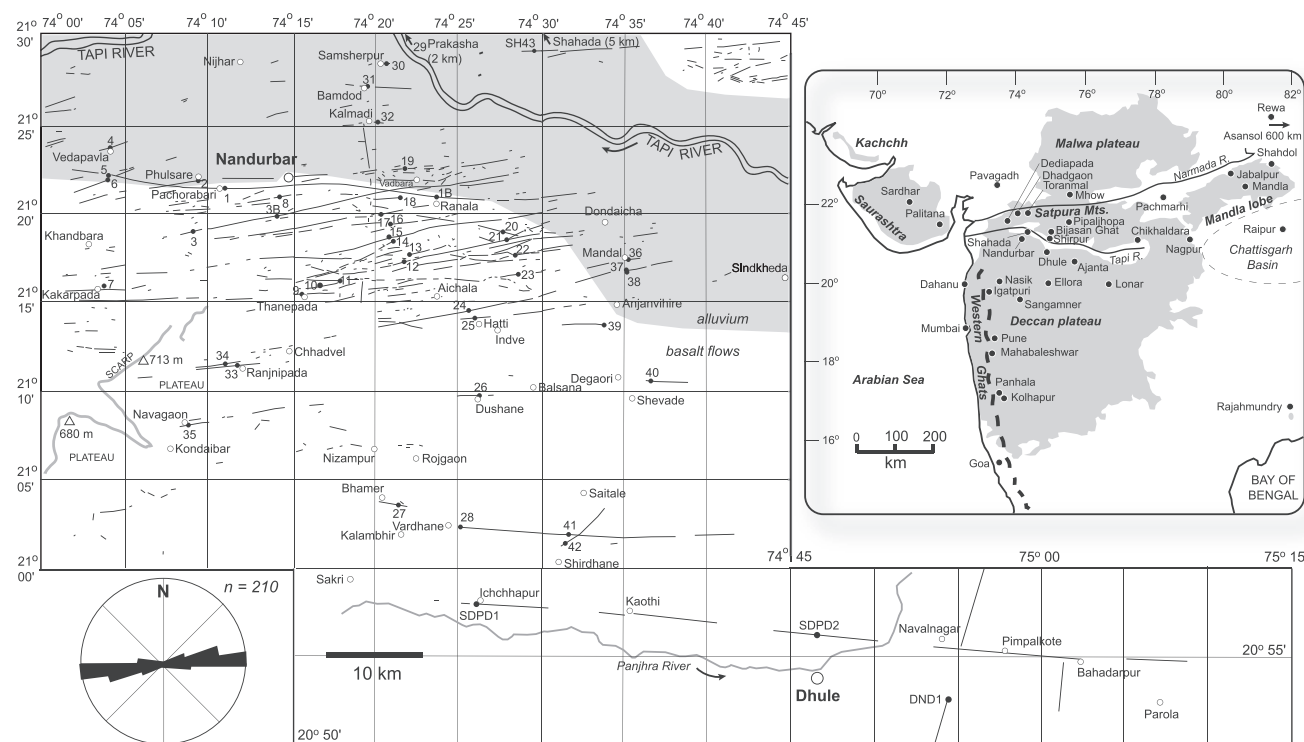


FIGURE 1 Map of the Nandurbar-Dhule dyke swarm (modified after Ray et al., 2007) showing the major physiographic and geological features, the dykes (black lines), and sample locations (black dots). All sample numbers have the prefix NBD (not shown). Towns and villages are shown with open circles. The alluvial belt of the Tapi River is shaded. At right is a key map showing the Deccan flood basalt province (shaded), its various regions, and localities mentioned in this paper as well as some others. The elliptical area south of the Mandla lobe shows the additional area of the Deccan Traps rocks that are mostly in the subsurface (after Chalapathi Rao et al., 2011). The main map includes a region covered by six survey of India toposheets of 1:50,000 scale. A rose diagram of the trends of 210 dykes (of >1 km length) measured over a larger region, covering 20 survey of India topographic sheets, is shown at bottom left (see Ray, Sheth, & Mallik, 2007 for details)

stratigraphically and petrogenetically unrelated tholeiitic sequences, such as Pavagadh and Palitana, suggest polycentric eruptions (e.g., Sheth & Melluso, 2008; Sheth, Zellmer, Kshirsagar, & Cucciniello, 2013). The Western Ghats sequence has recently been geochronologically bracketed by high-precision U-Pb ages (on zircon) of 66.288 ± 0.027 Ma for the Jawhar Formation (Fm.) at the base to 65.544 ± 0.015 Ma for the Mahabaleshwar Fm. (Table 1; Schoene et al., 2015). New $^{40}\text{Ar}/^{39}\text{Ar}$ results on plagioclase separates show generally consistent results throughout the Western Ghats sequence and an age of 66.38 ± 0.05 Ma for the Jawhar Fm. (Renne et al., 2015). In comparison, tholeiitic and alkalic rocks of the northern Deccan Traps (e.g., Pavagadh) have recently been dated at 66.6–64.9 Ma (Parisio et al., 2016).

Continental flood basalt provinces form from large, dyke-fed fissure eruptions (e.g., Ray, Sheth, & Mallik, 2007; Swanson, Wright, & Helz, 1975). The Deccan Traps contain three major tholeiitic dyke swarms (Auden, 1949; Deshmukh & Sehgal, 1988; Vanderkluyzen, Mahoney, Hooper, Sheth, & Ray, 2011). These are the ENE-WSW-trending Narmada-Tapi dyke swarm, in the region of the Narmada and Tapi rivers and the intervening Satpura mountains (e.g., Bhattacharji, Chatterjee, Wampler, Nayak, & Deshmukh, 1996; Ray et al., 2007; Sant & Karanth, 1990), the NNW-SSE-trending Coastal dyke swarm, on the Konkan plain between the western Indian coast and the Western Ghats escarpment (e.g., Dessai & Viegas, 1995; Hooper, Widdowson, & Kelley, 2010; Sheth et al., 2014; Widdowson, Pringle, & Fernandez, 2000), and the Nasik-Pune dyke swarm east of the Western Ghats escarpment, with more randomly oriented dykes

than the other two (e.g., Bondre, Hart, & Sheth, 2006; Vanderkluyzen et al., 2011). Abundant tholeiitic dykes also outcrop in the Saurashtra peninsula in the northwestern Deccan (Figure 1), forming a westerly extension of the Narmada-Tapi dyke swarm (Auden, 1949; Cucciniello, Demonerova, Sheth, Pande, & Vijayan, 2015; Sheth et al., 2013). Additional Deccan tholeiitic dykes outcrop in east-central India between the outcrop areas of the Deccan and the Early Cretaceous (~115 Ma) Rajmahal Traps (Chalapathi Rao et al., 2011; Kent, Saunders, Kempton, & Ghose, 1997). Geochemical (including Sr-Nd-Pb isotopic) studies on dykes of the three major swarms, sometimes aided by magnetic polarity criteria, indicate Deccan lava flows several hundred kilometres in length, and whereas the lower formations (Kalsubai Subgroup) of the Western Ghats sequence (Table 1) may have been fed by the Narmada-Tapi dyke swarm, the middle and upper formations (Lonavala and Wai subgroups) were fed by the Coastal and Nasik-Pune swarms (Bondre et al., 2006; Vanderkluyzen et al., 2011).

The well-developed Narmada-Tapi giant dyke swarm (Ernst & Buchan, 1997) is made up of several smaller mafic dyke swarms, such as the Dediapada, Dhadgaon, and Nandurbar-Dhule swarms in western India (Krishnamacharlu, 1971; Melluso, Sethna, Morra, Khateeb, & Javeri, 1999; Ray et al., 2007; Subbarao, Hooper, Dayal, Walsh, & Gopalan, 1999) and the Pachmarhi and Rewa-Shahdol swarms in central India (Lala, Choudhary, Patil, & Paul, 2011; Sheth et al., 2009). The Narmada-Tapi swarm is densest in the Dediapada and Nandurbar areas, with an average crustal extension of 5–6% (Krishnamacharlu, 1971; Ray et al., 2007). Vanderkluyzen et al. (2011) presented major

TABLE 1 Geochemical stratigraphy of the Western Ghats region, Deccan Traps

Group	Subgroup	Formation	Polarity	Age (Ma) $^{40}\text{Ar}/^{39}\text{Ar}$	U–Pb	$(^{87}\text{Sr}/^{86}\text{Sr})_t$	$\epsilon_{\text{Nd}}(t)$	$(^{143}\text{Nd}/^{144}\text{Nd})_t$	$^{206}\text{Pb}/^{204}\text{Pb}$
Deccan Basalt	Wai	Desur* (~100 m)	N			0.7072–0.7080	–5.9 to –10.4	0.512251–0.512020	na
		Panhala (>175 m)	N			0.7046–0.7055	+3.8 to –0.5	0.512748–0.512527	na
		Mahabaleshwar (280 m)	N	63.1 ± 1.0–69.7 ± 2.4	65.544 ± 0.015	0.7040–0.7055	+5.6 to –7.4	0.512840–0.512174	16.65–18.18
		Ambenali (500 m)	R	64.3 ± 0.8–66.4 ± 0.5	65.661 ± 0.087	0.7038–0.7044	+8.3 to +2.5	0.512978–0.512681	17.53–18.48
		Poladpur (375 m)	R	65.84 ± 0.58–68.04 ± 0.80		0.7053–0.7110	+3.3 to –11.0	0.512722–0.511989	17.40–19.32
	Lonavala	Bushe (325 m)	R			0.7078–0.7200	–7.5 to –19.4	0.512169–0.511559	18.33–22.85
		Khandala (140 m)	R			0.7071–0.7124	–1.0 to –20.4	0.512502–0.511507	16.69–19.26
		Bhimashankar (140 m)	R	67.23 ± 0.6		0.7067–0.7077	+0.2 to –5.9	0.512563–0.512251	19.60–20.92
	Kalsubai	Thakurvadi** (650 m)	R	67.5 ± 0.8–68.6 ± 0.6		0.7067–0.7224	–2.9 to –14.0	0.512404–0.511835	17.28–20.28
		Neral (100 m)	R	66.4 ± 1.0–68.0 ± 1.4		0.7062–0.7104	–2.5 to –15.0	0.512425–0.511784	16.68–19.95
		Jawhar-Igatpuri (>700 m)	R	66.5 ± 2.0–67.8 ± 0.6	66.288 ± 0.027	0.7085–0.7128	–3.2 to –8.5	0.512389–0.512117	19.12–22.52

Note. *The Desur is considered by some workers as a "Unit" of the Panhala Formation. **The Sr-isotopic range for most of the Thakurvadi Formation lavas is 0.7067–0.7112, but a single flow in the formation (Paten Basalt) has anomalous, broadly Bushe-like values of $(^{87}\text{Sr}/^{86}\text{Sr})_t = 0.7224$ and $\epsilon_{\text{Nd}}(t) = -15.4$. Formation names and thicknesses and isotopic ranges are based on Subbarao and Hooper (1988) and references therein, Lightfoot and Hawkesworth (1988), Lightfoot et al. (1990), Peng, Mahoney, Hooper, Harris, and Beane (1994), and Vanderkluysen et al. (2011), including their new lava flow analyses). N = normal magnetic polarity, R = reverse magnetic polarity. Note that the R–N polarity transition (29R to 29N) occurs not at the Ambenali–Mahabaleshwar contact but a little above it (Jay, MacNicol, Widdowson, Self, & Turner, 2009). All Sr and Nd isotopic ratios are initial ratios age-corrected to 65 million years. Pb isotopic ratios are present-day values. na = not analysed. $^{40}\text{Ar}/^{39}\text{Ar}$ ages are significant $^{40}\text{Ar}/^{39}\text{Ar}$ plateau ages for samples from different formations (Richards et al., 2015) using the calibration of Renne et al. (2010) and Renne et al. (2011), which makes $^{40}\text{Ar}/^{39}\text{Ar}$ and U–Pb ages comparable. See Richards et al. (2015) for reference to publications with original data. U–Pb ages are from Schoene et al. (2015), including individual ages for Jawhar-Igatpuri and Ambenali formations, and weighted mean age for the Mahabaleshwar Formation.

and trace element data for several hundred dykes (and Sr–Nd–Pb isotopic data on 60 dykes) in the Narmada–Tapi, Coastal, and Nasik–Pune swarms. However, relatively few dykes of the Narmada–Tapi swarm have been analysed for isotopes (five from Dediapada, three from Dhadgaon, and five from Nandurbar–Dhule). Because these dense tholeiitic dyke swarms contain many possible feeder dykes of the Deccan flood basalts, we have now carried out a $^{40}\text{Ar}/^{39}\text{Ar}$ geochronological and Sr–Nd–Pb isotopic study of the Nandurbar–Dhule dyke swarm. Based on good $^{40}\text{Ar}/^{39}\text{Ar}$ ages obtained on three of the dykes, and isotopic and elemental data on 17 dykes, we discuss the emplacement age and tectonic implications of the Nandurbar–Dhule swarm, its role as a feeder dyke swarm for Deccan lavas, and its implications for stratigraphic development across the Deccan Traps flood basalt province.

2 | GEOLOGY AND PETROGRAPHY

The Satpura mountains, running across central India in an ENE–WSW direction (Figure 1), and with many peaks >1 km above sea level, separate the valleys of the west-flowing Narmada and Tapi rivers. The Nandurbar–Dhule area, in the Tapi valley, is flat and low-lying (200 m above sea level) and exposes highly weathered, horizontal to gently north-dipping, compound pāhoehoe basalt lavas (Ray et al., 2007), partly covered by alluvium (Figure 1). The Nandurbar–Dhule mafic dyke swarm cuts through Deccan basalts and may have fed lava flows that were locally eroded but remain preserved in sections in the Satpura Range and the Western Ghats. The dykes are vertical, range in trace length from <1 to 79 km (Figure 1) and in thickness from 3 to 62 m; some appear to have formed by multiple magma injections. The mean strike of 210 dykes >1 km in length is N88°, but several long (tens of km) dykes trend ENE–WSW and better reflect the direction of the time-averaged minimum compressive stress (σ_3). A 54-km-long dyke (NBD1) has a gently wavy trace with an overall N95° trend, whereas the longest (79 km) Sakri–Dhule–Parola dyke (SDPD) also trends N95° (Figure 1).

The Nandurbar–Dhule dykes are evolved tholeiitic basalts and basaltic andesites with MgO = 3.30–6.00 wt.% and both low-Ti and high-Ti characters, and these magmas experienced low-pressure equilibration and fractional crystallization (especially of olivine) in shallow-level crustal magma chambers (Melluso et al., 1999). Such chambers have also been inferred from gravity modelling (Bhattacharji, Sharma, & Chatterjee, 2004) and computations of dyke magma overpressures (Ray et al., 2007; see this paper for detailed field descriptions). The dykes are relatively fresh, medium- or fine-grained, and composed mainly of calcic plagioclase and augite. They also contain accessory olivine (sometimes altered) and Fe–Ti oxides. Dykes that appear aphyric in hand specimen often show, under the microscope, microphenocrysts of plagioclase, clinopyroxene, or olivine, within an ophitic or subophitic groundmass (Ray, 2009).

3 | $^{40}\text{Ar}/^{39}\text{Ar}$ GEOCHRONOLOGY

3.1 | Analytical methods

For $^{40}\text{Ar}/^{39}\text{Ar}$ dating, we selected the freshest whole-rock samples of four of the longest dykes (NBD3, 36 km; NBD5, 54 km; NBD21,

32 km; SDPD1, 79 km). Sample chips were crushed, washed with high-purity water, and sieved to 180–250 μm size fraction, from which plagioclase was hand-picked under a binocular microscope. Sample NBD21 is a plagioclase-megacryst basalt and from it, high-purity plagioclase was separated. In the other samples, however, plagioclase was always in intergrowth with other minerals and, thus, the samples are plagioclase-enriched groundmass. All samples were pretreated with 4 N HNO_3 for 10 min at room temperature to ensure that any secondary minerals were completely removed. The treated samples were wrapped in high-purity aluminium foil and irradiated at the WWR-K reactor in Tomsk (Russia) together with the BERN4 muscovite monitor sample. Irradiation conditions were the same as reported in Vernikovskiy et al. (2003). Corrections for Ca- and K-interfering reactions were made using the following ratios derived from irradiated K_2SO_4 and calcium CaF_2 : $(^{36}\text{Ar}/^{37}\text{Ar})_{\text{Ca}} = 0.000446 \pm 0.000004$, $(^{39}\text{Ar}/^{37}\text{Ar})_{\text{Ca}} = 0.000891 \pm 0.000003$, and $(^{40}\text{Ar}/^{39}\text{Ar})_{\text{K}} = 0.089 \pm 0.001$. The conversion factor for the derivation of Ca/K ratio from the $^{37}\text{Ar}/^{39}\text{Ar}$ ratio was 3.7. The neutron flux gradient and related variations in the J factor were assessed via measurements of nine BERN4 muscovite age standards placed from the top to bottom through the ampoule containing the samples of the dykes. For the J factor calculations, we took the conventional decay constants (Steiger & Jäger, 1977) and age of BERN4 muscovite to be 18.885 Ma (corresponding to the Fish Canyon (FC) sanidine standard with age of 28.305 ± 0.036 Ma after the calibration of Renne, Balco, Ludwig, Mundil, & Min, 2011; Renne, Mundil, Balco, Min, & Ludwig, 2010), which allows direct comparison with U–Pb ages. For comparison to the $^{40}\text{Ar}/^{39}\text{Ar}$ ages obtained relative to an earlier calibration of the FC sanidine with an age of 28.02 Ma, the age of the BERN4 muscovite has to be set to 18.7 Ma (Ivanov, He, Yang, Nikolaeva, & Paleskii, 2009). The difference between the two calibrations is $\sim 1\%$, or ~ 0.65 Myr on the absolute timescale, for the studied samples.

The BERN4 muscovite monitor sample consists of two mineral phases and shows disturbed argon-release age patterns, especially at size fractions $< 0.6 \mu\text{m}$ (Heri, Robyr, & Villa, 2014). However, our stepwise-heating experiments on the BERN4 muscovite revealed excellent plateau with minor radiogenic argon loss during the low temperature steps. Inhomogeneity of age monitors is a common problem in $^{40}\text{Ar}/^{39}\text{Ar}$ dating. There is no “perfect” age monitor; even the most commonly used sanidine monitors, such as FC and Alder Creek sanidines, show notable degrees of disturbance, sometimes in excess of 1% (Jicha, Singer, & Sobol, 2016; Phillips & Matchan, 2013). Our experience using the BERN4 muscovite monitor shows that the repetitive analyses of coeval samples or different aliquots of the same sample of an a priori known age, which were irradiated in different runs with the same or different age monitors, always agree within 2% and in most cases within 1% of their age.

The argon isotopic measurements were made at the Centre for Geodynamics and Geochronology, Institute of the Earth's Crust of the Siberian Branch of the Russian Academy of Sciences, Irkutsk, Russia. We used a Thermo Scientific ARGUS VI multicollector mass spectrometer with a double-vacuum resistance furnace that allows heating the sample to more than 1,700 $^{\circ}\text{C}$. Temperature was measured via a thermocouple calibrated through optical pyrometer measurements, the melting point of aluminium (660 $^{\circ}\text{C}$), and room temperature. Accuracy of temperature measurements is thought to be better than

± 20 $^{\circ}\text{C}$ at high temperatures and ± 2 $^{\circ}\text{C}$ at room temperature. Instrumental mass discrimination was assessed via measurements of atmospheric argon with the $^{40}\text{Ar}/^{36}\text{Ar}$ set at 298.56 (Lee et al., 2006) and a power law for calculation of the discrimination (Renne, Cassata, & Morgan, 2009).

3.2 | Results

The $^{40}\text{Ar}/^{39}\text{Ar}$ dating results are summarized in Table 2 and the apparent-age argon release plots for the dyke samples are shown in Figure 2. Samples NBD3, NBD5, and SDPD1 yielded very good plateau spectra (Figure 2a–c). We define a plateau as composing a minimum of 60% of the total ^{39}Ar released and four or more successive degassing steps whose mean ages overlap at the 2σ level, not including the error contribution from the J value. The fourth sample, NBD21, did not yield a plateau spectrum though it was a high-purity plagioclase separate (Figure 2d). Its Ca/K ratio of ~ 60 suggests that it was low-K plagioclase, typical of tholeiitic rocks. The four highest temperature steps for this sample yield inverse and normal isochron ages with high errors (Table 2) and are thus not discussed further, although the analytical data are available from the Supporting Information.

The plateau, isochron, and inverse isochron ages for dykes NBD3, NBD5, and SDPD1 are concordant, with large amounts of total released ^{39}Ar (67.3% to 95.8%) for the plateau steps, acceptable mean square weighted deviate values of the isochrons and inverse isochrons, and atmospheric $^{40}\text{Ar}/^{36}\text{Ar}$ values of the trapped argon given by their intercepts (Table 2). Ca/K varies in these plateau steps significantly due to the fact that the samples were plagioclase-enriched matrix and other lower- and higher-K phases also contributed to the signal (Figure 2a–c). Despite the Ca/K variation, the good plateaus suggest that each sample was fresh and yielded its true crystallization age. The Sakri-Dhule-Parola dyke SDPD1 has an age of 67.06 ± 0.60 Ma, the Nandurbar dyke NBD5 has an age of 67.49 ± 0.89 Ma, and the Nandurbar dyke NBD3 has an age of 63.43 ± 0.48 Ma (2σ errors including the error contribution from the J value).

4 | GEOCHEMISTRY

4.1 | Analytical methods

We analysed 12 of the Nandurbar-Dhule dykes for Sr and Nd isotopic ratios at the Centre for Geodynamics and Geochronology, Institute of the Earth's Crust of the Siberian Branch of the Russian Academy of Sciences, Irkutsk, Russia. Sample preparation protocols followed Pin, Briot, Bassin, and Poitrasson (1994) for Sr and Pin and Santos-Zalduegui (1997) for Nd, with minor modifications outlined by Rasskazov, Saranina, Demonterova, Maslovskaya, and Ivanov (2002) and Ivanov, Demonterova, Rasskazov, and Yasnygina (2008). Isotopic ratios were measured using a Finnigan MAT262 thermal ionization mass spectrometer. Accuracy of the analyses was tested by measurements on the reference materials SRM 987 (0.710263 ± 0.000011 , 2σ , $n = 10$) and JNd-1 (0.512102 ± 0.000005 , 2σ , $n = 12$). Sr–Nd isotopic data for the 12 dyke samples analysed, and for five samples (NBD1,

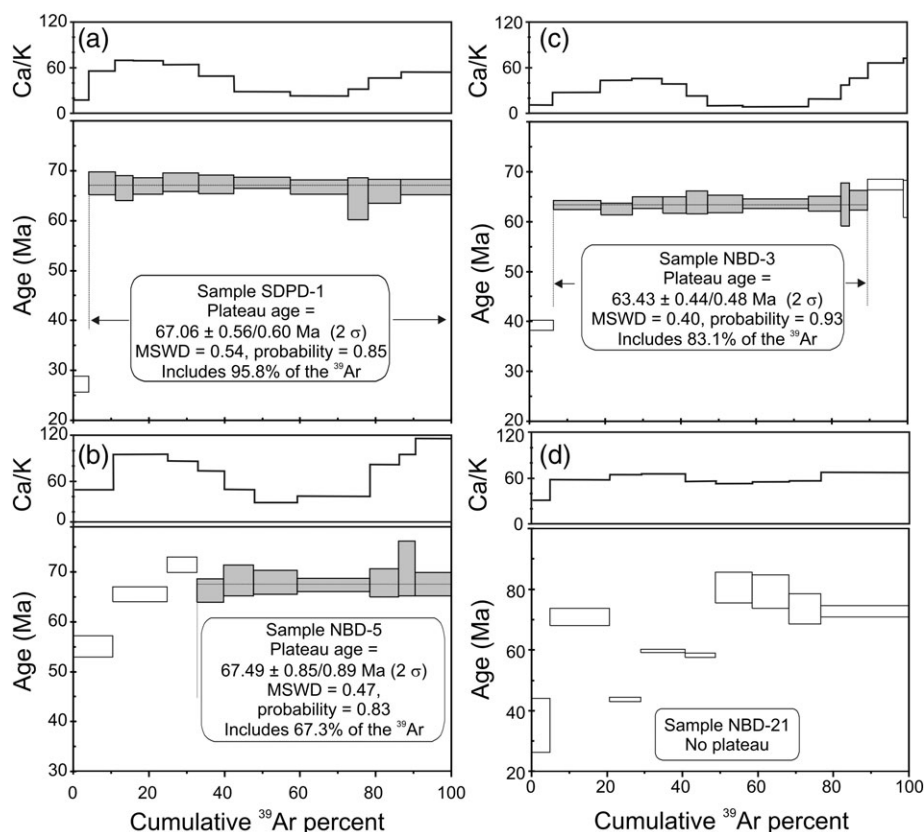


FIGURE 2 $^{40}\text{Ar}/^{39}\text{Ar}$ plateau spectra for the Nandurbar-Dhule dyke samples (a) SDPD1, (b) NBD5, (c) NBD3, and (d) NBD21. The plateau steps are shown with grey shading and the non-plateau steps are in white. Also shown are the % ^{39}Ar released and values of the MSWD (mean square weighted deviate) and probability (p). Variations in Ca/K ratio for each sample are also shown

TABLE 3 Sr-Nd isotopic data for the Nandurbar-Dhule dyke swarm

Trend	Length (km)	Sample	$^{87}\text{Rb}/^{86}\text{Sr}$	$(^{87}\text{Sr}/^{86}\text{Sr})_p$	$\pm 2\sigma$	$(^{87}\text{Sr}/^{86}\text{Sr})_t$	$^{147}\text{Sm}/^{144}\text{Nd}$	$(^{143}\text{Nd}/^{144}\text{Nd})_p$	$\pm 2\sigma$	$(^{143}\text{Nd}/^{144}\text{Nd})_t$	ϵNd_t
N85	53	NBD1	0.331	0.715260	0.000016	0.714954	0.134	0.511859	0.000009	0.511802	-14.7
N75	36	NBD3	0.215	0.704905	0.000014	0.704707	0.158	0.512739	0.000008	0.512672	+2.3
N65	3	NBD4	0.510	0.710070	0.000016	0.709599	0.159	0.512310	0.000009	0.512242	-6.1
N95	54	NBD5	0.423	0.715017	0.000012	0.714626	0.121	0.511907	0.000008	0.511856	-13.6
N60	24	NBD6	0.184	0.711880	0.000013	0.711710	0.136	0.511967	0.000009	0.511909	-12.6
N80	2.5	NBD10	0.318	0.711190	0.000016	0.710897	0.147	0.512360	0.000009	0.512297	-5.0
N80	24	NBD15	0.231	0.704983	0.000009	0.704770	0.153	0.512654	0.000007	0.512589	+0.7
N83	22	NBD17	0.208	0.704964	0.000014	0.704772	0.162	0.512672	0.000007	0.512603	+1.0
N85	24	NBD19	0.149	0.704678	0.000014	0.704541	0.145	0.512697	0.000007	0.512636	+1.6
N70	32	NBD21	0.190	0.704565	0.000013	0.704389	0.158	0.512793	0.000006	0.512726	+3.4
N75	9	NBD22	0.200	0.705011	0.000009	0.704827	0.158	0.512742	0.000008	0.512675	+2.4
N85	6.7	NBD31	0.797	0.725677	0.000014	0.724941	0.136	0.511612	0.000007	0.511554	-19.5
N82	6.7	NBD33	0.175	0.704585	0.000014	0.704424	0.159	0.512727	0.000007	0.512660	+2.1
N95	79	SDPD1	0.247	0.704524	0.000015	0.704296	0.158	0.512777	0.000007	0.512710	+3.0
N95	79	SDPD2	0.156	0.711687	0.000015	0.711543	0.150	0.512252	0.000012	0.512188	-7.1
~E-W	?	TAP1	0.510	0.70639	0.000016	0.70592	0.148	0.512682	0.000009	0.512619	+1.2
~E-W	?	TAP2	0.132	0.70473	0.000016	0.70461	0.163	0.512748	0.000009	0.512679	+2.4

Note. All reported dyke lengths are minimum lengths, owing to possibly incomplete exposure (see Ray et al., 2007). The NBD and SDPD samples are those of Ray et al. (2007). Samples TAP1 and TAP2 are those of Melluso et al. (1999), with field data not available. Subscripts “p” and “t” on Sr and Nd isotopic ratios and ϵNd values indicate present-day and initial (age-corrected to 65 million years) values, respectively. Age-corrections to isotopic ratios using parent-daughter concentrations measured on small sample chips by isotope dilution (where available), and bulk-rock concentrations measured by ICPMS, are generally very close. Analytical uncertainty on ϵNd is ± 0.1 – 0.2 units. Data for dykes NBD1, NBD4, NBD10, TAP1, and TAP2 are from Vanderkluyzen et al. (2011) and all remaining data were collected in the present study, on unleached samples. Question marks indicate that this information is not available.

TABLE 4 Pb isotopic data for the Nandurbar-Dhule dyke swarm

Sample	$^{206}\text{Pb}/^{204}\text{Pb}$	$\pm 2\sigma$	$^{207}\text{Pb}/^{204}\text{Pb}$	$\pm 2\sigma$	$^{208}\text{Pb}/^{204}\text{Pb}$	$\pm 2\sigma$
NBD1	18.883	0.011	15.800	0.012	39.090	0.012
NBD3	17.591	0.007	15.406	0.008	38.175	0.020
NBD4	20.031	0.011	15.863	0.012	40.640	0.012
NBD5	17.591	0.007	15.408	0.008	38.179	0.021
NBD6	18.825	0.008	15.743	0.009	38.968	0.021
NBD10	22.360	0.011	16.085	0.012	43.163	0.012
NBD15	18.302	0.007	15.617	0.008	39.097	0.021
NBD17	17.341	0.007	15.381	0.008	38.111	0.020
NBD19	17.759	0.007	15.427	0.008	38.248	0.020
NBD21	18.046	0.007	15.458	0.008	38.461	0.020
NBD22	17.490	0.007	15.384	0.008	38.095	0.020
NBD31	20.142	0.008	15.449	0.008	39.878	0.021
NBD33	17.729	0.007	15.414	0.009	38.302	0.021
SDPD1	17.730	0.007	15.405	0.008	38.251	0.021
SDPD2	19.108	0.008	15.719	0.009	40.082	0.021
TAP1	17.637	0.011	15.556	0.012	37.639	0.012
TAP2	17.764	0.011	15.415	0.012	38.411	0.012

Note. All Pb isotopic ratios are present-day values, following the norm in Deccan studies. Data for dykes NBD1, NBD4, NBD10, TAP1, and TAP2 are from Vanderkluyzen et al. (2011) and all remaining data were collected in the present study, on unleached samples. Analytical details, standards, and errors are presented in the text.

Bushe-like flow in the Thakurvadi Fm. (Vanderkluyzen et al., 2011; see Figure 3 and Table 1).

Other Nandurbar-Dhule dykes also show substantially radiogenic Sr and unradiogenic Nd isotopic compositions. These include dyke NBD4 located in the Thakurvadi Fm. field but close to the Khandala, Poladpur, and Jawhar-Igatpuri Fm. fields and dyke NBD6 located near but just outside the Thakurvadi Fm. field. Other examples are dykes NBD10 and SDPD2 located within the Jawhar-Igatpuri Fm. field. Dyke TAP1 is located in the long, narrow field of the crustally contaminated Poladpur Fm., but near its high- $\epsilon_{\text{Nd}}(t)$, relatively uncontaminated end. There is a distinct clustering of Sr-Nd isotopic data near the high- $\epsilon_{\text{Nd}}(t)$ end of the mixing arrays representing primarily mantle-derived, “depleted” compositions, in the area of overlap between the Mahabaleshwar-Panhala and Ambenali Fms. (By “depleted,” we mean representing long-term depletion in the more incompatible trace elements). Data for as many as nine of the 17 isotopically analysed Nandurbar-Dhule dyke samples (NBD3, NBD15, NBD17, NBD19, NBD21, NBD22, NBD33, SDPD1, and TAP2) plot in this restricted area with $\epsilon_{\text{Nd}}(t)$ from +3.4 to +0.7 and ($^{87}\text{Sr}/^{86}\text{Sr}$)_i from 0.70430 to 0.70483.

The formation affinities of the Nandurbar-Dhule dykes identified with the Sr-Nd plot (Figure 3) are generally all validated by the Nd-Pb and Sr-Pb plots (Figure 4a,b). For example, dykes NBD1, NBD5, NBD6, and NBD31, and the Boradi dyke SH49, are all broadly Bushe-like or Thakurvadi-like. Dyke NBD10 is located within or very near the Jawhar-Igatpuri Fm., as in the Sr-Nd isotopic plot, but dyke SDPD2 is not, being located within or just outside the Thakurvadi Fm. field, somewhat far from the Jawhar-Igatpuri field. Still, the nine dykes that are Mahabaleshwar-type in Sr-Nd isotopic ratios are located well within or very near the small Mahabaleshwar Fm. field in Figure 4a,b.

5 | GEOCHEMICAL CORRELATIONS TO DECCAN LAVAS

5.1 | Lateral geochemical homogeneity and heterogeneity in dykes

Before we discuss the geochemical correlations of individual dykes to the lava flows of the Deccan Traps, it is necessary to evaluate how homogeneous in chemical-isotopic composition the individual dykes are. Examples of regional Deccan dykes showing lateral homogeneity or heterogeneity have been discussed by Bondre et al. (2006), Ray et al. (2007), Sheth et al. (2009), Sheth et al. (2013), Vanderkluyzen et al. (2011), and Cucciniello et al. (2015). The 79-km-long Sakri-Dhule-Parola dyke shows very contrasting compositions in samples SDPD1 (isotopically Mahabaleshwar-like, with $\epsilon_{\text{Nd}}(t) = +3.0$, Zr/Nb = 12.23, Ba/Y = 3.05) and the much more highly crustally contaminated SDPD2 (Jawhar-Igatpuri-like, with $\epsilon_{\text{Nd}}(t) = -7.1$, Zr/Nb = 17.37, Ba/Y = 5.87). Keshav, Sheth, and Chandrasekharam (1998) physically traced this dyke over 40 km around Dhule and noted that it was composed of two separate magma injections, not always well exposed. Samples SDPD1 and SDPD2 thus may not represent lateral heterogeneity in a single dyke but rather two injections (at least) of compositionally very distinct magma batches (Figures 3, 4) into the dyke fissure.

An additional example of a regional dyke with lateral heterogeneity is the 54-km-long dyke from which samples NBD5, NBD1, and NBD1B come (Figure 1). Samples NBD5 (with $\epsilon_{\text{Nd}}(t) = -13.6$, Zr/Nb = 12.73, Ba/Y = 9.37), and NBD1 (with $\epsilon_{\text{Nd}}(t) = -14.7$, Zr/Nb = 17.43, Ba/Y = 9.50) sampled 10 km eastwards, are somewhat different in isotopic (including Pb) and chemical composition, and though both are broadly Bushe-like (Figure 5a), NBD1 arguably represents a greater degree of contamination by ancient basement crust, and these differences between NBD5 and NBD1 can be produced by increasing contamination of a single magma batch flowing eastwards. However, sample NBD1B taken 20 km east of NBD1 has Zr/Nb = 17.3 and Ba/Y = 9.50 (isotopes are not available; Vanderkluyzen et al., 2011), values that are essentially those of NBD1, and therefore, the two samples may represent a single, homogeneous magma batch.

West to east magma transport with increasing crustal contamination is thus suggested by the data available for both long dykes. Ray et al. (2007) gave reasons to consider Nandurbar-Dhule dykes with lengths of tens of kilometres as formed by lateral (rather than vertical) injection. They hypothesized that systematic geochemical changes along strike may occur in them, noting that the laterally injected Jurassic dykes of eastern North America (Greenough & Hodych, 1990) and the 1.27 Ga Mackenzie giant dykes of Canada (Baragar, Ernst, Hulbert, & Peterson, 1996) show greater crustal contamination downstream. It is clear that the Nandurbar-Dhule dykes cannot be assumed to be internally homogeneous; the correlations we achieve below should thus be considered correlations between lava flows and dykes, or specific segments of dykes.

5.2 | Using geochemical data for dyke-flow correlations

Geochemical data have been used successfully in stratigraphic correlation of widely separated sections in the Deccan Traps, and include Sr-

TABLE 5 Major and trace element data for the Nandurbar-Dhule dyke swarm

Lat.	21°21' 30"	21°19' 00"	21°23' 50"	21°22' 30"	21°22' 25"	21°22' 45"	21°18' 45"	21°20' 00"	21°22' 30"	21°18' 45"	21°17' 35"	21°27' 30"	21°11' 30"	20°58' 00"	20°56' 30"	—	—
Long.	74°10' 50"	74°08' 45"	74°03' 45"	74°03' 45"	74°03' 45"	74°03' 45"	74°20' 50"	74°20' 05"	74°21' 45"	74°27' 35"	74°28' 00"	74°19' 30"	74°11' 30"	74°26' 00"	74°46' 30"	—	—
Trend	N85°	N75°	N65°	N95°	N60°	N60°	N80°	N83°	N85°	N70°	N75°	N85°	N82°	N95°	N95°	-E-W	-E-W
Length (km)	53	36	3	54	24	24	24	22	24	32	9	6.7	6.7	79	79	?	?
Width (m)	5.4	10	3.8	18	?	?	11	20	7	5.5	?	10	20	7	6.5	?	?
DFA	—	Ner	Bush	Kha	Tha	Tha	—	Kha/Pol	Bhi	Mah	Jaw-Ig	Bush	Bhi	Jaw-Ig	Kha	?	?
Sample	NBD1	NBD3	NBD4	NBD5	NBD6	NBD10	NBD15	NBD17	NBD19	NBD21	NBD22	NBD31	NBD33	SDPD1	SDPD2	TAP1	TAP2
SiO ₂	52.43	49.60	51.95	52.34	51.65	50.53	51.35	50.00	50.10	49.21	49.10	53.45	49.79	49.74	52.58	49.90	50.54
TiO ₂	0.93	2.60	1.30	0.91	0.92	2.53	2.18	2.72	2.54	3.34	3.04	0.78	3.02	3.13	1.98	2.28	2.72
Al ₂ O ₃	15.35	13.60	14.37	15.71	16.46	14.10	16.33	13.06	14.41	13.28	13.56	15.56	13.60	12.90	14.77	13.84	14.93
FeO _T	10.61	14.30	11.91	10.32	9.72	13.61	11.43	14.79	13.26	15.51	14.18	9.88	14.09	15.44	12.40	13.78	12.50
MnO	0.18	0.22	0.19	0.17	0.16	0.20	0.18	0.23	0.20	0.22	0.21	0.16	0.21	0.22	0.19	0.22	0.18
MgO	6.92	6.54	6.78	6.82	6.68	5.75	4.40	6.17	5.88	5.40	6.30	6.63	5.77	5.27	4.94	5.29	4.60
CaO	10.85	10.28	10.17	10.99	11.62	9.60	10.50	10.09	10.71	9.85	10.63	10.33	10.39	9.92	10.12	9.05	9.72
Na ₂ O	1.82	2.03	1.83	1.72	1.97	2.40	2.60	2.09	2.18	2.28	2.12	2.01	2.27	2.11	2.29	2.38	2.51
K ₂ O	0.77	0.55	1.34	0.88	0.69	0.98	0.70	0.54	0.47	0.56	0.53	1.10	0.54	0.93	0.53	1.45	0.43
P ₂ O ₅	0.14	0.29	0.14	0.14	0.13	0.30	0.32	0.30	0.25	0.34	0.32	0.10	0.30	0.33	0.21	0.31	0.47
Mg#	57.8	48.9	54.4	58.1	59.0	47.0	44.7	46.7	48.2	42.2	48.2	58.4	46.2	41.7	45.5	44.0	43.0
Sc	40	40	44	32	38	31	28	45	32	38	41	38	38	37	32	40	25
Cr	94	92	143	155	169	67	101	88	195	95	83	199	146	71	72	284	170
Co	42	49	45	38	42	41	31	50	39	50	49	42	47	49	42	—	—
Ni	57	77	71	120	66	38	42	67	84	69	82	90	83	56	44	117	92
Cu	105	268	167	103	85	93	223	265	186	306	236	89	259	267	126	225	192
Zn	67	129	87	69	69	102	102	136	113	158	127	60	140	158	101	128	110
Rb	20	15	34	31	16	29	27	14	15	15	16	38	14	19	13	33	13
Sr	165	202	135	212	252	264	338	195	292	228	232	138	232	223	241	181	268
Y	24	39	33	24	18	31	40	41	36	41	37	23	39	42	32	41	31
Zr	129	169	117	84	92	137	190	186	163	183	177	91	203	208	172	151	152
Nb	7.4	15	8.9	6.6	7.1	13	15	14	14	20	13	5.8	17	17	9.9	17	15
Cs	0.24	0.22	0.20	0.19	0.11	0.85	0.20	0.15	0.13	0.20	0.13	0.53	0.17	0.18	0.76	—	—
Ba	228	141	144	225	193	204	187	138	120	130	133	154	121	128	188	268	166
La	15	15	12	16	12	20	19	14	16	17	13	14	16	16	17	16	13
Ce	30	36	25	32	24	44	44	35	38	41	33	28	38	39	39	45	30
Pr	3.7	5.1	3.2	3.7	3.0	5.8	5.8	4.9	5.1	5.7	4.6	3.2	5.3	5.6	5.0	—	—
Nd	15	23	13	14	12	25	26	22	23	26	21	12	24	26	21	—	—

(Continues)

TABLE 5 (Continued)

Sm	3.2	6.0	3.5	2.8	2.7	6.1	6.6	5.9	5.5	6.8	5.5	2.7	6.3	6.8	5.2	—	—
Eu	1.0	1.8	1.1	1.0	1.0	1.8	2.1	1.8	1.7	2.1	1.7	0.8	1.9	2.1	1.6	—	—
Gd	3.8	6.4	4.2	3.4	3.2	6.2	7.0	6.2	5.8	7.0	5.8	3.2	6.4	7.0	5.5	—	—
Tb	0.63	1.1	0.76	0.54	0.51	0.97	1.1	1.1	1.0	1.1	1.0	0.55	1.1	1.2	0.91	—	—
Dy	3.8	6.4	4.9	3.5	2.9	5.4	6.9	6.5	5.8	6.8	6.0	3.4	6.4	7.1	5.4	—	—
Ho	0.82	1.3	1.1	0.77	0.62	1.0	1.4	1.3	1.2	1.3	1.2	0.73	1.2	1.4	1.1	—	—
Er	2.4	3.6	3.2	2.1	1.7	2.7	3.5	3.6	3.1	3.5	3.3	2.2	3.4	3.8	2.9	—	—
Tm	0.38	0.55	0.51	0.33	0.27	0.39	0.50	0.56	0.43	0.53	0.50	0.35	0.51	0.56	0.44	—	—
Yb	2.3	3.2	3.1	2.1	1.7	2.3	3.3	3.3	2.6	3.0	2.9	2.1	3.0	3.2	2.6	—	—
Lu	0.36	0.48	0.47	0.35	0.25	0.33	0.52	0.49	0.42	0.44	0.43	0.33	0.43	0.48	0.38	—	—
Hf	3.1	4.3	2.9	2.1	2.3	3.7	4.9	4.8	4.2	4.7	4.5	2.3	5.1	5.3	4.4	—	—
Ta	0.48	0.96	0.57	0.39	0.39	0.85	0.84	0.90	0.84	1.3	0.86	0.41	1.1	1.1	0.63	—	—
Pb	6.1	2.0	2.7	5.3	2.8	2.3	2.8	2.1	1.9	1.8	1.9	5.2	2.0	1.7	4.2	—	—
Th	3.0	1.8	2.9	2.6	1.6	3.9	2.1	1.8	1.8	1.8	1.7	5.2	1.9	1.8	3.9	—	—
U	0.79	0.43	0.74	0.66	0.36	0.81	0.50	0.41	0.41	0.40	0.40	1.2	0.44	5.0	6.0	—	—

Note. Data for NBD samples are from Vanderkluyzen et al. (2011) and for TAP samples from Melluso et al. (1999), who reported total iron as $\text{Fe}_2\text{O}_{3\text{T}} = 15.31$ and 13.89 wt.% for TAP1 and TAP2, respectively. For these samples, $\text{FeO}_{\text{T}} = 0.8998 * \text{Fe}_2\text{O}_{3\text{T}}$. Mg Number ($\text{Mg\#}$) = $100 \text{ Mg}^{2+}/(\text{Mg}^{2+} + \text{Fe}^{2+})$, atomic, assuming 85% of the total Fe to be in the Fe^{2+} form. Many more analyses can be found in both papers; this table only lists those samples that are isotopically analysed (this study and Vanderkluyzen et al., 2011). Dashes for concentrations mean elements not analysed, and dashes for DFA mean no discriminant function analysis matches. Question marks mean information not available.

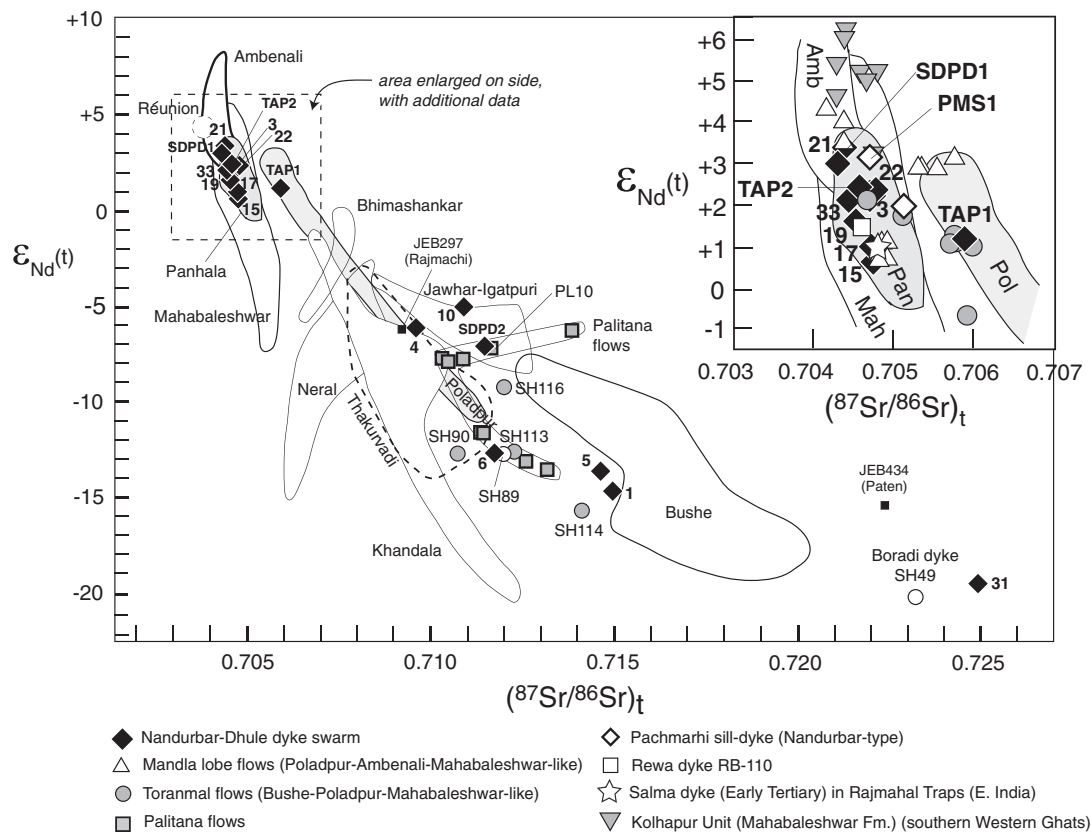


FIGURE 3 Sr-Nd isotopic plot for the Nandurbar-Dhule dykes, with sample numbers indicated in boldface (without the prefixes to avoid cluttering). Data are shown for the Western Ghats stratigraphic formations (Vanderkluyzen et al., 2011, including their new lava flow analyses). A close-up of the boxed area with dashed boundaries is shown in the inset figure, with some additional data not plotted in the main figure such as for some Toranmal and Mandla lobe lava flows. The isotopic ratios are initial ratios for 65 Ma. Analytical errors are much smaller than the sizes of the symbols. Data sources are as follows: Nandurbar-Dhule dykes (this study and Vanderkluyzen et al., 2011), Mandla lobe flows (Shrivastava, Mahoney, & Kashyap, 2014), Toranmal flows and dykes (Mahoney et al., 2000), Shahada-Shirpur area (including Boradi) dykes (Chandrasekharam, Mahoney, Sheth, & Duncan, 1999), Palitana flows (Sheth et al., 2013), Pachmarhi dyke and Chakhla-Delakhari sill (Sheth et al., 2009), Rewa-Shahdol dyke RB-110 (Lala et al., 2011), Salma dyke in the Rajmahal Traps (Kent et al., 1997), Kolhapur unit of the southernmost Deccan (Lightfoot et al., 1990), and the Rajmachi and Paten Basalt members of the Western Ghats sequence (Vanderkluyzen et al., 2011)

Nd-Pb isotopes (not modified by fractional crystallization) and alteration-resistant trace elements (thus excluding Rb and K, and also Ba and Sr for highly weathered samples; e.g., Beane et al., 1986; Chandrasekharam et al., 1999; Devey & Lightfoot, 1986; Lightfoot et al., 1990; Lightfoot & Hawkesworth, 1988; Mahoney et al., 2000; Peng et al., 1998; Peng et al., 2014; Sheth, Mahoney, & Chandrasekharam, 2004). The same data have been used in evaluating dyke-flow correlations (e.g., Bondre et al., 2006; Cucciniello et al., 2015; Sheth et al., 2009; Sheth et al., 2013; Vanderkluyzen et al., 2011). Lavas of many Western Ghats formations show significant overlap in binary plots, particularly of the major elements, but they show distinctive characteristics of primitive mantle-normalized multi-element patterns (e.g., Mahoney et al., 2000; Peng et al., 1994; Sheth et al., 2004; Sheth et al., 2009). Guided by the isotopic matches between individual Nandurbar-Dhule dykes and various Western Ghats formations (Figures 3, 4), we found many close matches in their multi-element patterns (Figure 5a–g) and in binary plots of alteration-resistant incompatible elements and their ratios (Figure 6). An additional tool that has been applied in geochemical correlations is discriminant function analysis (DFA); however, DFA does not well discriminate formations with similar concentrations of

key elements and should not be used independently of the isotopic and alteration-resistant elemental criteria (see Bondre et al., 2006; Peng et al., 1998; Peng et al., 2014; Sheth et al., 2004; Sheth et al., 2009; Vanderkluyzen et al., 2011).

5.3 | Partial geochemical matches between Nandurbar-Dhule dykes and Deccan flows

Isotopic and elemental matches between dykes and flows frequently do not go hand in hand: Dyke-flow pairs with close isotopic similarities can be significantly different in their elemental characteristics, and dykes and flows with essentially the same elemental characteristics may have isotopic ratios that preclude a correlation and genetic relationship. The latter situation is exemplified by the Rewa-Shahdol dyke RB-110 of Lala et al. (2011) and lava flow PLBF12/S12 of Shrivastava, Mahoney, and Kashyap (2014). The dyke (with Zr/Nb = 13.85, Zr/Y = 4.99, and Ti/Y = 413) has a Mahabaleshwar- and Panhala-like Sr-Nd isotopic composition, with $(^{87}\text{Sr}/^{86}\text{Sr})_t = 0.70462$ and $\epsilon_{\text{Nd}}(t) = +1.5$. The flow, with nearly the same Zr/Nb (13.85), Zr/Y (4.39), and Ti/Y (414), is isotopically Ambenali-like, with $(^{87}\text{Sr}/^{86}\text{Sr})_t = 0.70417$ and $\epsilon_{\text{Nd}}(t) = +4.3$; note that

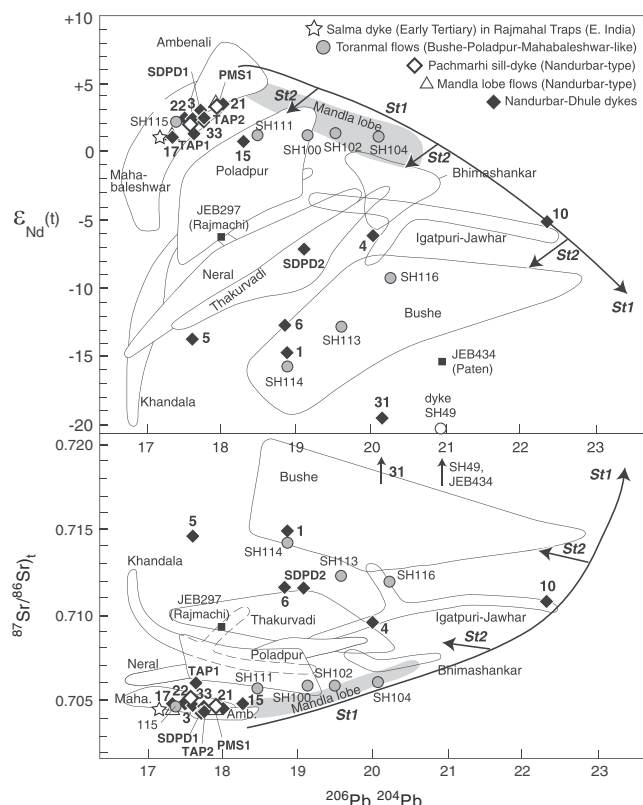


FIGURE 4 (a, b) Nd-Pb and Sr-Pb isotopic plots for the Nandurbar-Dhule dykes, with sample numbers indicated in boldface (without the prefixes to avoid cluttering). Data are shown for the Western Ghats stratigraphic formations (Vanderkluyzen et al., 2011, including their new lava flow analyses). The Pb isotopic ratios are present-day values, in consistency with previous Deccan studies. Analytical errors are much smaller than the sizes of the symbols. Note that two sample points for the Mandla lobe flows (open triangles) are hidden behind other sample symbols because of compositional overlap. Pb isotopic data are not available for Palitana flows and Rewa-Shahdol dyke RB-110. Data sources are as in Figure 3. Heavy arrows in both panels indicated by St1 and St2 schematically illustrate the two-stage model of contamination proposed by Peng et al. (1994), and the high- $^{206}\text{Pb}/^{204}\text{Pb}$ flows around Jabalpur are shown by the grey-shaded field labelled “Mandla lobe”

neither the isotopic ratios nor these highly incompatible element ratios are sensitive to varying degrees of fractional crystallization.

Similarly, there are good chemical equivalents of the Nandurbar-Dhule dykes in lava flows of the Palitana section in Saurashtra but without isotopic matches. Sheth et al. (2013, see their figure 11) showed the close similarity in normalized multi-element patterns of the uppermost three, high-Ti Palitana flows, and the high-Ti Nandurbar dyke NBD10 ($\text{TiO}_2 = 2.53$ wt.%, Nb = 13 ppm, Zr = 137 ppm, Y = 31 ppm), but also noted that $(^{87}\text{Sr}/^{86}\text{Sr})_t$ of flows PL1 and PL3 (Sheth et al., 2013) and dyke NBD10 (Vanderkluyzen et al., 2011) were, respectively, 0.71050, 0.71029, and 0.71093, and thus different. The identical $\epsilon_{\text{Nd}}(t)$ values of flows PL1 and PL3 (−8.0 and −7.8) were also different from that of the dyke NBD10 (−5.1), and thus, this dyke could not have fed these flows. Sheth et al. (2013) noted that all these four samples chemically matched the Sakri-Dhule-Parola dyke SDPD1 ($\text{TiO}_2 = 3.13$ wt.%, Nb = 17 ppm, Zr = 208 ppm, and Y = 42 ppm), and so this dyke was a potential

feeder, pending verification from isotopic data. Our isotopic data for SDPD1 ($(^{87}\text{Sr}/^{86}\text{Sr})_t = 0.70430$, $\epsilon_{\text{Nd}}(t) = +3.0$) now indicate that there is no possible genetic relationship between dyke SDPD1 and these chemically very similar Palitana flows.

In the Sr-Nd isotopic plot (Figure 3), data for dyke NBD4 plot at the edge of the Thakurvadi Fm. field but also near the Khandala, Poladpur, and Jawhar-Igatpuri Fm. fields. The dyke's isotopic composition ($(^{87}\text{Sr}/^{86}\text{Sr})_t = 0.70960$, $\epsilon_{\text{Nd}}(t) = -6.1$) is closely similar to that of the Rajmachi flow in the Khandala Fm. analysed by Vanderkluyzen et al. (2011; sample JEB-207, $(^{87}\text{Sr}/^{86}\text{Sr})_t = 0.70925$, $\epsilon_{\text{Nd}}(t) = -6.2$). The $^{206}\text{Pb}/^{204}\text{Pb}$, $^{207}\text{Pb}/^{204}\text{Pb}$, and $^{208}\text{Pb}/^{204}\text{Pb}$ ratios of the Rajmachi member (18.015, 15.545, and 39.281, respectively) are much lower than those of the dyke NBD4 (Figure 4a,b). The multi-element pattern of the Rajmachi flow is rather similar to the NBD4 pattern but with a difference in the slopes of the two for the heavy rare earth elements (HREE; Figure 5c).

Data for dyke NBD6 plot between the Thakurvadi and Bushe Fm. fields in Figures 3, 4a,b. NBD6, with $(^{87}\text{Sr}/^{86}\text{Sr})_t = 0.71171$ and $\epsilon_{\text{Nd}}(t) = -12.6$, nearly overlaps with Toranmal dyke SH89 (0.71199, −12.8) and Toranmal flow SH113 (0.71228, −12.6; Mahoney et al., 2000) but has lower Pb isotopic ratios. Chemically also, dykes NBD6 and SH89 and flow SH113 are much more Bushe-like than Thakurvadi-like (Figure 5d). Sr-Nd isotopic data for dykes NBD10 and SDPD2 plot well within the Jawhar-Igatpuri Fm. field (Figure 3). However, in the Nd-Pb and Sr-Pb isotopic plots (Figure 4a,b), whereas the NBD10 data plot within or at the edge of the Jawhar-Igatpuri Fm. field, the SDPD2 data plot somewhat far from that field and within or very near the Thakurvadi Fm. field, thus precluding a complete correspondence for SDPD2 with Jawhar-Igatpuri characteristics. Yet both dykes have normalized multi-element patterns closely resembling those of flows of the Jawhar-Igatpuri Fms. (Figure 5e,f).

Numerous examples of this kind show that the isotopic and chemical variability in the Western Ghats sequence, albeit very large, is only part of the entire isotopic and chemical range of the Deccan flood basalts (tholeiitic flows and dykes). Whereas many of the stratigraphic formations in the Western Ghats are well separated from each other in isotopic or elemental plots, the data for all Deccan tholeiites taken together would show much greater continuity.

5.4 | Broad (formation-level) isotopic and chemical matches

In the Sr-Nd-Pb isotopic plots (Figures 3, 4), data for dykes NBD1 and NBD5 plot very close to the Bushe Fm. field and data for dyke NBD31 locate it farther than the Bushe field. Dykes and flows of broadly Bushe-like isotopic and chemical compositions are known from various sections in the Malwa plateau and Satpura Range, including Mhow-Pipaljhopa (Peng et al., 1998; Subbarao et al., 1999), Toranmal (Mahoney et al., 2000), and Bijasan Ghat (Sheth et al., 2004), as well as small sections in the Shahada-Shirpur area in the Tapi valley (Chandrasekharam et al., 1999). There are close matches in the primitive mantle-normalized multi-element patterns of the Nandurbar dykes NBD1, NBD5, and NBD31, the Boradi dyke SH49, the Pingalvadi member, and the Bushe Chemical Type (both from the Bushe Fm.), and the broadly Bushe-like Toranmal flows SH114 and SH116

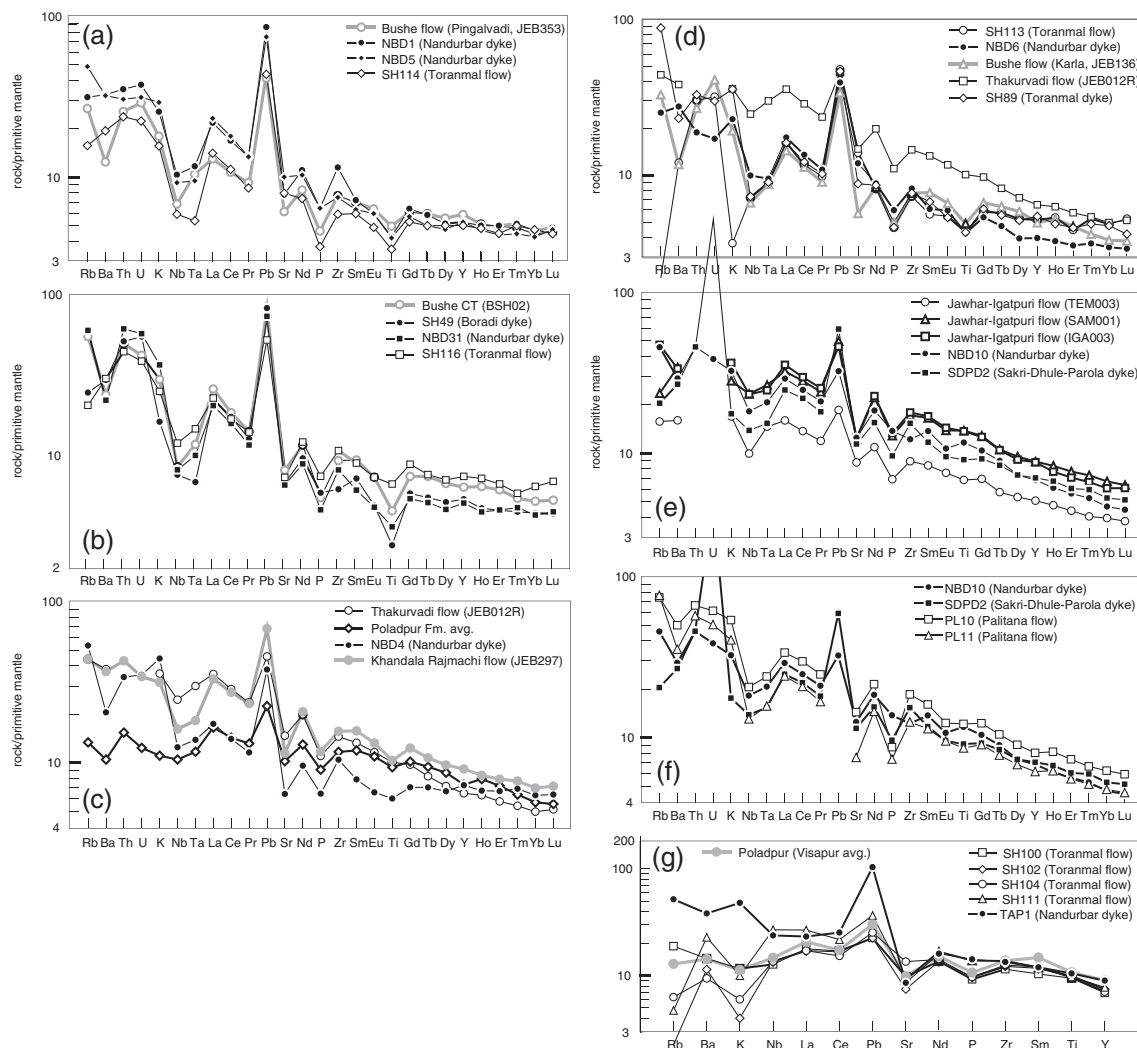


FIGURE 5 Comparison of primitive mantle-normalized multi-element patterns of the Nandurbar-Dhule dykes with those of closely similar Western Ghats lavas, members, or formation averages (main data sources are Beane et al., 1986; Beane, 1988; Vanderkluyzen et al., 2011). The normalizing values are from Sun and McDonough (1989). (a, b) Broadly Bushe-type compositions, (c, d) Mixed compositions, (e, f) Jawhar-Igatpuri-type compositions, and (g) Poladpur-type compositions

(Mahoney et al., 2000; Figure 5a,b). Peng et al. (1998), Subbarao et al. (1999), Mahoney et al. (2000), and Sheth et al. (2004) have noted that the broadly Bushe-like flows in the northern sections, which notably do not obey the formation-stratigraphic order seen in the Western Ghats, are unrelated to the Bushe Fm. of the Western Ghats and probably erupted from local feeder dykes. None of the Nandurbar-Dhule dykes has isotopic composition exactly or closely like any of these broadly Bushe-like northern Deccan flows, so these dyke-formation geochemical similarities are at a broad level.

5.5 | Close isotopic and chemical matches at both formation and individual flow levels

Our new isotopic data, combined with existing isotopic and elemental data (Melluso et al., 1999; Vanderkluyzen et al., 2011), indicate possible feeder dykes in the Nandurbar-Dhule dyke swarm, with matches not only at the larger, formation level but also at the finer, individual flow level.

The isotopic composition of dyke SDPD2 (with $(^{87}\text{Sr}/^{86}\text{Sr})_t = 0.71154$ and $\epsilon_{\text{Nd}}(t) = -7.1$) completely overlaps with that of lava flow PL10 in the Palitana section (with $(^{87}\text{Sr}/^{86}\text{Sr})_t = 0.71168$ and $\epsilon_{\text{Nd}}(t) = -7.2$). None of

the Palitana basalts have been analysed for Pb isotopes given their highly weathered nature. The field of the Palitana flows in Sr-Nd isotopic space (Figure 3) is distinctly boomerang-shaped, primarily because of a possible weathering alteration effect on lava flow PL11 (loss on ignition 7.00 wt.%) increasing its $^{87}\text{Sr}/^{86}\text{Sr}$ ratio, while leaving the Nd isotopic ratio unaffected (Sheth et al., 2013). In that case, flow PL11 also may correspond to the Jawhar-Igatpuri Fm. In terms of multi-element patterns, Palitana flows PL10 (sampled 335 m above sea level) and PL11 (310 masl) are closely similar to the dykes NBD10 and SDPD2 and the Jawhar-Igatpuri Fm. lavas (Figure 5e,f). The flows' elevations are consistent with the Jawhar Fm. having its upper boundary at 400 m in the Thalghat type section near Igatpuri (Beane et al., 1986). We think that these Palitana flows represent a northwesterly extension of the Jawhar Fm. into Saurashtra. Thus, some Nandurbar-Dhule dykes fed some lavas of the Jawhar Fm. 200–250 km to the south-south-west in the Western Ghats and two flows in the Palitana section 200 km to the west-north-west. In particular, Palitana flow PL10 was likely fed by dyke SDPD2.

Dyke TAP1 ($(^{87}\text{Sr}/^{86}\text{Sr})_t = 0.70592$, $\epsilon_{\text{Nd}}(t) = +1.2$) analysed by Vanderkluyzen et al. (2011) plots well within and near the high- $\epsilon_{\text{Nd}}(t)$ end of the Poladpur Fm. field in Figure 3. The Toranmal section is

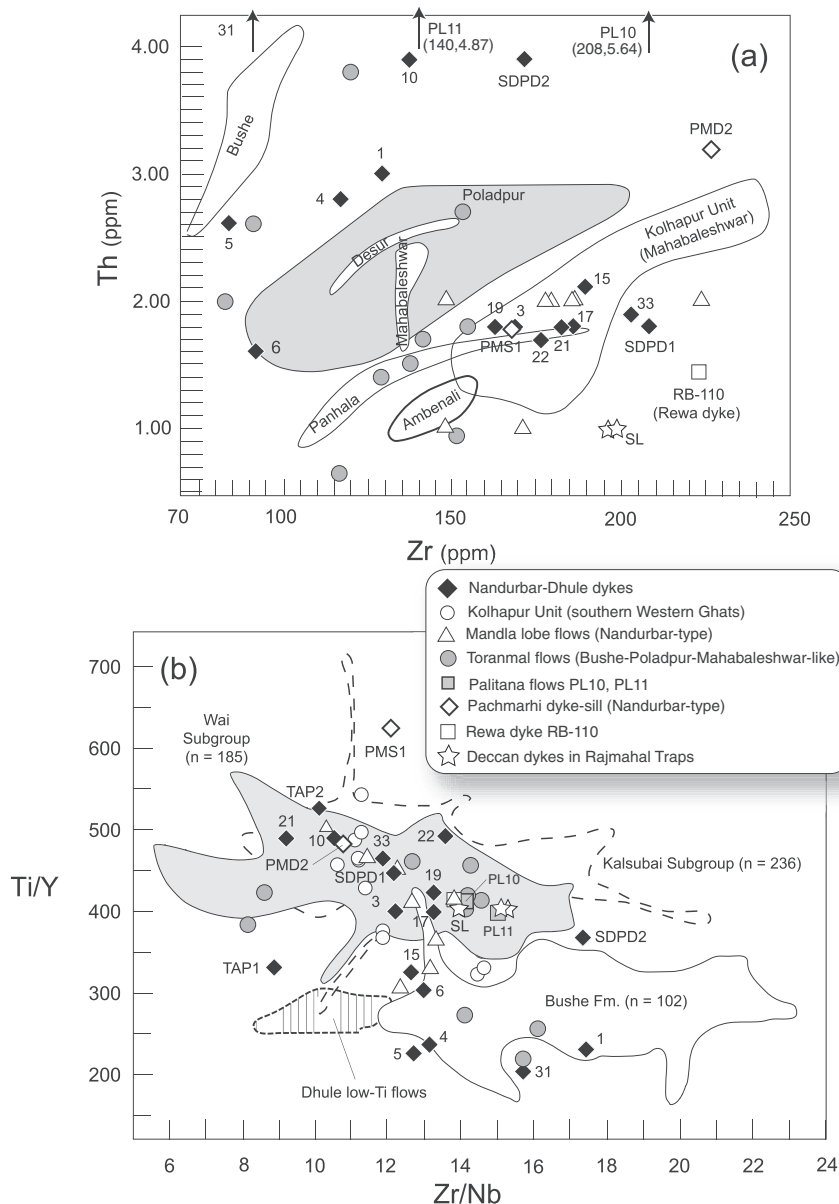


FIGURE 6 (a) Plot of Th versus Zr, showing the concentration ranges of the younger stratigraphic formations and constituent units of the southern Western Ghats region and also the Bushe Formation based on Lightfoot et al. (1990). Also shown are the data for the Nandurbar-Dhule dykes and the other dyke-sill intrusions and lava flows mentioned in this study. (b) Plot of Ti/Y versus Zr/Nb for various Deccan mafic lavas and the Nandurbar-Dhule dykes. Data sources are as follows: The Kalsubai and Wai subgroups and the Khandala and Bushe formations of the Lonavala Subgroup (total 624 samples): Beane (1988); Dhule area low-Ti flows: Melluso, Barbieri, and Beccaluva (2004); Nandurbar-Dhule dykes: This study and Vanderkluisen et al. (2011), and Melluso et al. (1999) for samples TAP1 and TAP2. Though there are more than 40 Nandurbar-Dhule dyke samples analysed for the major and trace elements (Vanderkluisen et al., 2011), the data plotted here are for only those dykes with isotopic data available (17 samples)

dominated by Poladpur-type flows (Mahoney et al., 2000), and the isotopic composition of dyke TAP1 overlaps with or is very close to the compositions of Toranmal flows SH100 (0.70582, +1.1), SH102 (0.70579, +1.3), SH104 (0.70599, +1.0), and SH111 (0.70571, +1.1; Figure 3 inset). Whereas these flows are significantly different from the dyke TAP1 in their Pb isotopic ratios (Figure 4a,b), the chemical similarities between dyke TAP1 and these four flows are also striking. Only a limited elemental dataset is available for TAP1 in Melluso et al. (1999), but Figure 5g shows their very similar multi-element patterns. The closest matches for TAP1 are flows SH100 and SH104, and given the very close correspondence of dyke TAP1 to these flows

in all respects, we wonder if their differences in Pb isotopic ratios might reflect alteration processes, or are primary. If the former, then dyke TAP1 can be considered a well-demonstrated example of a feeder dyke of one of these two lava flows outcropping in the Toranmal section 100 km to the north-north-east.

Other close individual dyke-flow matches exist. Dyke NBD6 ($(^{87}\text{Sr}/^{86}\text{Sr})_t = 0.71171$, $\epsilon_{\text{Nd}}(t) = -12.6$) and Toranmal flow SH113 ($(^{87}\text{Sr}/^{86}\text{Sr})_t = 0.71228$, $\epsilon_{\text{Nd}}(t) = -12.6$) are identical in Nd isotopic ratio and very close in Sr and Pb isotopic ratios, a small mismatch that may be explained by weathering (note that the loss on ignition value for SH113, at 5.06 wt.%, is very high and likely caused by weathering).

Chemically, there is a fair degree of correspondence between NBD6 and SH113 (Figure 5d).

5.6 | A new chemical-isotopic flavour in the Deccan tholeiites

We noted that as many as nine of the 17 dykes now isotopically analysed from the Nandurbar-Dhule swarm have Sr-Nd isotopic compositions plotting in the area of overlap between the Mahabaleshwar and Panhala Fms. (Figure 3 inset). Most of them also plot well within the small Mahabaleshwar Fm. field in the Nd-Pb and Sr-Pb isotopic plots (Figure 4a,b). However, these nine dykes (NBD3, NBD15, NBD17, NBD19, NBD21, NBD22, NBD33, SDPD1, and TAP2) cannot represent the Panhala Fm., which has $\text{TiO}_2 < 2.3$ wt.% and $\text{Zr/Nb} > 13$ (Lightfoot & Hawkesworth, 1988; Table 6), because these dykes generally have $\text{TiO}_2 > 2.5$ wt.% (and up to 3.34 wt.%), with Zr/Nb mostly < 13 . These dykes also cannot be true Mahabaleshwar-type dykes, because their Zr/Nb ratios are mostly higher than the upper limit of 10.5 for Mahabaleshwar Fm. (Table 6). Besides, the young (63.5 Ma) age of NBD3 and the older (67.1 Ma) age of SDPD1 preclude a correlation with the Mahabaleshwar Fm. of the Western Ghats (mean age of 65.544 ± 0.015 Ma, Schoene et al., 2015).

Lightfoot et al. (1990) mapped a sequence of flows above the Ambenali Fm. in the Kolhapur area and the southernmost Western Ghats (Figure 1), where the Mahabaleshwar Fm. does not outcrop. They named this sequence the Kolhapur Unit, which is characterized by high TiO_2 (> 2.7 and up to 4.0 wt.%), low Sr (< 240 ppm), and low Zr/Nb (< 13). The nine Nandurbar-Dhule dykes mentioned above have high TiO_2 with Sr mostly < 240 ppm (181–232 ppm in seven samples and 292–338 ppm in two others) and Zr/Nb mostly < 13 (8.88–12.67 in seven samples and 13.28–13.61 in two others). Thus, chemically, these dykes are essentially identical to the Kolhapur Unit of the southernmost Deccan. However, they are generally dissimilar to the Kolhapur Unit in isotopic ratios, because the latter's isotopic values (Figure 3 inset) locate it at the more “depleted,” high- $\epsilon_{\text{Nd}}(t)$ end of the Mahabaleshwar Fm. field and well into the Ambenali Fm. field. Thus, these nine Nandurbar-Dhule dykes combine geochemical

features of three eruptive sequences in the Wai Subgroup: Sr-Nd-Pb isotopic ratios of the Mahabaleshwar and Panhala Fms. but not of the Kolhapur Unit, and chemical characteristics strongly like those of the Kolhapur Unit and unlike those of the Mahabaleshwar and Panhala Fms. This geochemical flavour is hitherto unknown in the Deccan tholeiites anywhere in the province, and we term it “Nandurbar-type.”

Figure 6a is a plot of Th versus Zr for the Nandurbar-Dhule dykes and the various other Deccan rock suites discussed in this paper, and Figure 6b is a plot of Ti/Y versus Zr/Nb ratios. All four elements are alteration-resistant. The Zr/Nb ratio is unchanged even during extreme alteration (e.g., Mitchell & Widdowson, 1991), and both ratios are insensitive to the olivine gabbro fractionation well recognized in the Western Ghats sequence (e.g., Cox & Hawkesworth, 1984, 1985; Cucciniello et al., 2014; Devey & Cox, 1987; Lightfoot et al., 1990). The two ratios however change in opposite directions during crustal contamination. The important point to note is that the data for six of the nine “Nandurbar-type” dykes recognized above plot well within the Kolhapur Unit field, and two others are located very close to the field (Th concentration is not available for the ninth dyke TAP1). None of these data plot anywhere near the Mahabaleshwar Fm. field, and several other dyke data (e.g., for the broadly Bush-like dykes NBD1, NBD5, and NBD31) plot near the fields of formations to which they are isotopically or otherwise similar. In the Ti/Y versus Zr/Nb plot (Figure 6b), data for the Kolhapur Unit lavas and the “Nandurbar-type” dykes plot close together. The close chemical similarity of these two is thus a robust feature.

Data for Deccan tholeiitic intrusions in central and eastern India also generally plot near these two suites (Figure 6a,b). These include the Chakhla-Delakhari sill (PMS1) and a Pachmarhi dyke (PMD2; Sheth et al., 2009), Rewa-Shahdol dyke RB-110 (Lala et al., 2011), Deccan tholeiite dykes in the Rajmahal Traps (Kent et al., 1997), and several Mandla lobe lava flows (Shrivastava et al., 2014).

Figure 7a shows the close matches between multi-element patterns of the isotopically Mahabaleshwar-type Toranmal flow SH115 ($\text{TiO}_2 = 2.74$ wt.%, $(^{87}\text{Sr}/^{86}\text{Sr})_t = 0.70467$, $\epsilon_{\text{Nd}}(t) = +2.1$, $^{206}\text{Pb}/^{204}\text{Pb} = 17.393$) with those of the dykes NBD3 (0.70471, +2.3, 17.591), NBD22 (0.70483, +2.4, 17.490), and NBD33 (0.70442,

TABLE 6 Geochemical criteria used to distinguish the younger stratigraphic units, Western Ghats

Formation	Sr (ppm)	Ba (ppm)	Ba/Y	$(^{87}\text{Sr}/^{86}\text{Sr})_t$	Zr/Nb	TiO_2 (wt.%)
Desur Unit (Panhala)	> 230	> 150	—	0.707–0.708	< 12.5	Low (< 2.25)
Panhala	< 200	< 90	—	0.704–0.705	> 13	Low (< 2.2)
Mahabaleshwar	> 250	> 100	> 4	> 0.705	< 10.5	> 2.0
Kolhapur Unit	> 200	—	—	0.704–0.705	< 13.0	High (> 2.25)
Ambenali	200–250	< 100	< 3.5	< 0.705	10.5–15	< 2.7
Poladpur	—	> 100	> 3.5	0.705–0.713	15–20	—
Bushe	—	> 100	—	> 0.713	> 20	Very low (< 1.5)

Note. Based on Devey and Lightfoot (1986), Lightfoot et al. (1990), Mitchell and Widdowson (1991), and Jay and Widdowson (2008). The Kolhapur Unit directly overlies the Ambenali Fm. in the southernmost Deccan Traps (from Kolhapur southwards, Figure 1) and comprises highly fractionated flows with distinct geochemical features from the Mahabaleshwar Fm. as shown. The Desur lavas are considered a Unit of the Panhala Fm. by some workers; they overlie and are more radiogenic than the Panhala flows in the southernmost Deccan Traps but have trace element abundances similar to the Mahabaleshwar Fm. (e.g., Lightfoot et al., 1990). Dashes mean that the particular criterion is not defined for that particular formation or unit, or if defined, is not distinctive. The “Nandurbar-type” compositions defined in this study combine the Nd-Sr-Pb isotopic characteristics of the Mahabaleshwar and Panhala Fms. with the chemical characteristics of the Kolhapur Unit. These “Nandurbar-type” dykes mostly show Zr/Nb between 10.5 (upper limit for the Mahabaleshwar Fm.) and 13.0 (lower limit for the Panhala Fm.). Like the Kolhapur Unit (TiO_2 2.7–4.0 wt.%, Sr < 240 ppm, $\text{Zr/Nb} < 13$), the “Nandurbar-type” dykes mostly have $\text{TiO}_2 > 2.5$ wt.% (up to 3.34 wt.%), Sr < 240 ppm, and $\text{Zr/Nb} < 13$.

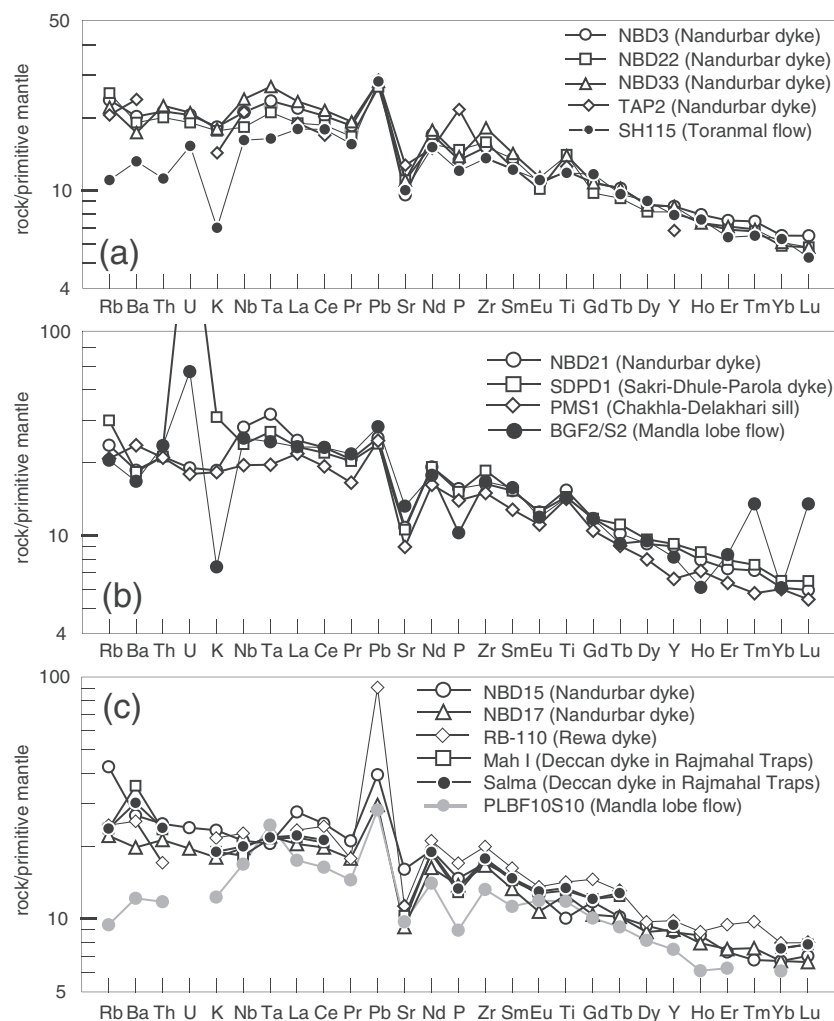


FIGURE 7 (a–c) Comparison of primitive mantle-normalized multi-element patterns of the Nandurbar-Dhule dykes with those of closely similar northern and northeastern Deccan lavas and intrusions, and Deccan-age dykes in the Rajmahal Traps of eastern India. The normalizing values are from Sun and McDonough (1989)

+2.1, 17.729). Dyke TAP2 (0.70461, +2.4), for which only a small trace element dataset exists (Melluso et al., 1999), is also similar but dykes NBD3 and NBD22 are especially similar to Toranmal flow SH115 in all respects including all isotopic ratios, and one of them is arguably the feeder dyke of that flow.

Figure 7b shows the closely similar multi-element patterns of the Nandurbar-Dhule dykes NBD21 ($(^{87}\text{Sr}/^{86}\text{Sr})_t = 0.70439$, $\epsilon_{\text{Nd}}(t) = +3.4$) and SDPD1 (0.70430, +3.0), and the sample PMS1 (0.70472, +3.1) representing the lower chilled margin of the Chakhla-Delakhari sill near Pachmarhi (Sheth et al., 2009). PMS1 has high TiO_2 of 3.13 wt.% and $\text{Zr}/\text{Nb} = 12.1$, perfectly within the range of the Kolhapur Unit and these “Nandurbar-type” dykes. In fact, the isotopic ratios of PMS1 completely overlap with those of the southern Deccan Kolhapur Unit flow BAS281 (0.70472, +3.2, Lightfoot et al., 1990; Figure 3 inset). Remarkably, the sill sample PMS1 and the dyke NBD21 are not only identical in Nd isotopes and very close in Sr isotopic ratios but also extremely close in Pb isotopic ratios (Figure 4a,b; $^{206}\text{Pb}/^{204}\text{Pb} = 17.939$ for PMS1 versus 18.046 for NBD21, $^{207}\text{Pb}/^{204}\text{Pb} = 15.440$ versus 15.458, and $^{208}\text{Pb}/^{204}\text{Pb} = 38.487$ versus 38.461). These rocks are closely similar to the pattern of the lava flow BGF2/S2 from the middle part of the Mandla lobe stratigraphy, with $(^{87}\text{Sr}/^{86}\text{Sr})_t = 0.70442$, $\epsilon_{\text{Nd}}(t) = +3.5$, $\text{TiO}_2 = 3.06$ wt.%, and $\text{Zr}/\text{Nb} = 10.4$ (Shrivastava et al., 2014). Note that the large peaks and troughs for the HREE in the normalized pattern of BGF2/S2 (Figure 7b) are an artefact of the low precision

of the HREE measurements as reported in Shrivastava et al. (2014). Significantly, like the other key geochemical parameters, the Pb isotopic ratios of BGF2/S2 ($^{206}\text{Pb}/^{204}\text{Pb} = 17.950$, $^{207}\text{Pb}/^{204}\text{Pb} = 15.450$, and $^{208}\text{Pb}/^{204}\text{Pb} = 38.523$) overlap with those of PMS1 (Figure 4a,b) and the two have identical Nd isotopic ratios within analytical error and very similar Sr isotopic ratios.

Figure 7c shows the close chemical similarity between “Nandurbar-type” dykes NBD15 and NBD17, Rewa-Shahdol dyke RB-110 (Lala et al., 2011, no Pb isotopes available), and Deccan dykes of Early Palaeocene age (Salma and Mah I) found cutting through the Rajmahal Traps of eastern India (Kent et al., 1997), and another Mandla lobe flow PLBF10S10 from near the top of the lava stratigraphy (Shrivastava et al., 2014). Their very close Sr–Nd isotopic values are as follows: NBD15 ($(^{87}\text{Sr}/^{86}\text{Sr})_t = 0.70477$, $\epsilon_{\text{Nd}}(t) = +0.7$) and NBD17 (0.70477, +1.0), RB-110 (0.70462, +1.5), Salma dyke (0.70482, +1.0), and PLBF10S10 (0.70494, +1.1). The closest match seems to be between PLBF10S10 and NBD17, and their Pb isotopic ratios are in agreement (Figure 4a,b), with $^{206}\text{Pb}/^{204}\text{Pb} = 17.348$ versus 17.341, $^{207}\text{Pb}/^{204}\text{Pb} = 15.355$ versus 15.381, and $^{208}\text{Pb}/^{204}\text{Pb} = 38.074$ versus 38.111; the Toranmal flow SH115 is also similar isotopically (Figures 3, 4). Their TiO_2 and Zr/Nb values are as follows: NBD15 (2.18 wt.%, 12.67), NBD17 (2.72 wt.%, 13.28), RB-110 (3.05 wt.%, 13.85), Salma (2.91 wt.%, 14.0), and PLBF10S10 (2.49 wt.%, 12.33).

We have shown that there are “Nandurbar-type” (isotopically Mahabaleshwar-Panhala-type, chemically Kolhapur-type) lava flows in the upper parts of the Mandla lobe (Shrivastava et al., 2014), which geochemically closely resemble several Nandurbar-Dhule dykes. DFA by Shrivastava et al. (2014) yielded a Poladpur Fm. match for them. Similarly, DFA carried out by Vanderkluyzen et al. (2011) yielded a “Neral > Khandala” match for the “Nandurbar-type” dyke NBD3 (see Table 5). The DFA results are thus in conflict with the isotopic and elemental matches we have achieved here, and the latter must be given more credence than the DFA, for reasons already noted.

6 | DISCUSSION

Our geochronological and geochemical data on the Nandurbar-Dhule dyke swarm provide insights into the evolutionary history of the Deccan Traps, including polycentric eruptions, duration, and tectonics of dyke swarm emplacement, and the stratigraphic development of different parts of the province.

6.1 | Polycentric eruptions

We inferred that Nandurbar-Dhule dykes such as NBD10 (and perhaps SDPD2) may have fed some Jawhar-Igatpuri Fm. lavas in the Western Ghats ~200 km to the south. The same or similar dykes also fed flows PL10 and PL11 in the Palitana section, which would therefore represent a northwesterly extension of the Jawhar Fm. into Saurashtra. If so, the 200 m exposed basalt thickness (five flows) below flows PL10-11 in the Palitana section (Sheth et al., 2013) would represent lava eruptions older than the Jawhar Fm. Thus, single sections such as Palitana, Toranmal (Mahoney et al., 2000), or Pavagadh (Sheth & Melluso, 2008) may not represent eruptive products of a single magma chamber or feeder dyke system. Instead, each was likely constructed with lava flows originating in separate eruptive areas, flowing various distances in different directions, and becoming juxtaposed, a scenario of polycentric eruptions. The junctions between these lava flows arriving from different eruptive centres, well seen in younger provinces such as the Miocene Columbia River flood basalt province (Reidel, Camp, Tolan, & Martin, 2013), have been removed by the much greater erosion that the Deccan Traps have undergone.

6.2 | Duration and tectonics of dyke swarm emplacement

The three high-quality $^{40}\text{Ar}/^{39}\text{Ar}$ ages of 67.06 ± 0.60 , 67.49 ± 0.89 , and 63.43 ± 0.48 Ma (2σ) that we obtained for the Nandurbar-Dhule dyke swarm show that the swarm emplacement occurred during 4.06 ± 0.96 Myr (which is the difference in the ages of the NBD3 and NBD5 dykes) and there likely are many old (~67 Ma) and young (~63 Ma) dykes in the swarm. Other radio-isotopic dates of dykes in the Narmada-Satpura-Tapi region are few but 66.9–64.5 Ma dyke ages (K-Ar and $^{40}\text{Ar}/^{39}\text{Ar}$) reported by Bhattacharji et al. (1996) and a 65.6 Ma ($^{40}\text{Ar}/^{39}\text{Ar}$) age by Sheth, Duncan, Chandrasekharam, and Mahoney (1997) fall within the time period indicated by our new dating results. A ~3–4 Myr minimum duration of dyke emplacement

is also known for the Central Saurashtra mafic dyke swarm (which includes the 62 km long Sardhar dyke; Cucciniello et al., 2015).

The obtained age span for the Nandurbar-Dhule dykes implies repetitive (or continuous) crustal dilation in the Nandurbar-Dhule area under a broadly uniform regional σ_3 direction (~N-S for the swarm with the N88° mean trend). There are both early (SDPD1, 67.06 ± 0.60 Ma) and late (NBD3, 63.43 ± 0.48 Ma) “Nandurbar-type” dykes, and they exhibit no preferred orientation but a 25° range of strike directions from N70° (NBD21), N75° (NBD3, NBD22), N80–82–83–85° (NBD15, 33,17,19) to N95° (SDPD1, NBD5).

6.3 | Independent evolution of the Western Ghats and the northern-northeastern Deccan Traps

As seen, broadly Bushe-like (both chemically and isotopically) lavas and dykes, now known from many areas in the northwestern, central, and northern Deccan, such as Saurashtra (Melluso et al., 1995; Sheth et al., 2013), Dhule (Melluso et al., 2004), Boradi near Shirpur (Chandrasekharam et al., 1999), Toranmal (Mahoney et al., 2000), Bijasan Ghat (Sheth et al., 2004), and Mhow-Pipaljhapa (Peng et al., 1998; Subbarao et al., 1999), cannot represent the Bushe Fm. of the Western Ghats. The Bushe chemical magma type of the Western Ghats is thus not chronostratigraphic but instead must have erupted at different times in various regions. This is similar to observations in the Paraná and Siberian flood basalt provinces (Ivanov et al., 2013; Peate, 1997).

The best evidence that the same magma type erupted at widely different times in the Deccan Traps is the ≥ 2.5 Myr difference in age between the early and late dykes with the same “Nandurbar-type” geochemistry, namely, SDPD1 (67.06 ± 0.60 Ma) and NBD3 (63.43 ± 0.48 Ma). For the geochemically closely similar Chakhla-Delakhari sill in central India, Sen and Cohen (1994) reported $^{40}\text{Ar}/^{39}\text{Ar}$ ages of 65.5 ± 0.3 and 66.1 ± 0.3 Ma (2σ), which fall neatly within the time interval between the dykes SDPD1 and NBD3. Kent et al. (1997) and Kent, Pringle, Muller, Saunders, and Ghose (2002) presented geochemical data and an $^{40}\text{Ar}/^{39}\text{Ar}$ age of 65.4 ± 0.6 Ma (2σ) on the Salma dyke and two associated, probable Deccan-age dykes found near Asansol, eastern India, in the area of the Rajmahal Traps. We have shown that these Deccan dykes Salma and Mah-I have multi-element patterns essentially identical to those of the “Nandurbar-type” Nandurbar-Dhule dykes, the Chakhla-Delakhari sill, and the Rewa-Shahdol dyke RB-110, as well as two Mandla lobe lavas (Figure 7c), all of which are closely similar in Sr-Nd-Pb isotopic ratios (Figures 3, 4), as well as emplacement age.

In contradistinction, the “Nandurbar-type” geochemistry is not found in the Western Ghats, which points to a geological evolution of the northern and northeastern Deccan independent from that of the Western Ghats. Also, Mandla lobe lavas found around Jabalpur, which otherwise resemble the Wai Subgroup of the Western Ghats sequence, have systematically higher Pb isotopic ratios than the Wai Subgroup (Figure 4a,b) and therefore are postulated to have erupted from different feeder dykes than the Wai Subgroup (Peng et al., 1998). The feeder dykes of the high- $^{206}\text{Pb}/^{204}\text{Pb}$ Jabalpur lavas remain unknown; no Nandurbar-Dhule dykes are of this type but dyke DD1 in the Dediapada swarm north-west of the Nandurbar-Dhule swarm has

these characteristics (Vanderkluyzen et al., 2011). Jabalpur is 600 km east-north-east of Dediapada, but we note that (a) dyke-flow correlations in the Deccan Traps suggest 150–500 km, and in some cases up to 700 km, of surface transport for several lava flows (Bondre et al., 2006; Sheth et al., 2009; Vanderkluyzen et al., 2011), (b) individual flows in the Columbia River province reached lengths >500 km and were only stopped by the Pacific Ocean (Reidel et al., 2013), and (c) 1,000 km of surface transport from the Western Ghats to the southeastern Indian coast has been suggested for the Rajahmundry basalts (Baksi, Byerly, Chan, & Farrar, 1994; Self, Jay, Widdowson, & Keszthelyi, 2008).

Vanderkluyzen et al. (2011) proposed that the Nandurbar-Dhule and associated dyke swarms (Dediapada, Dhadgaon) could have fed lavas of the lower and middle formations in the Western Ghats sequence, under ~N-S-directed crustal extension, whereas the upper formations were fed by dykes of the Coastal and Nasik-Pune swarms, the latter showing no preferred extension. Our geochemical data support Vanderkluyzen et al.'s (2011) proposal that some of the lower Western Ghats lavas could have originated in the Narmada-Tapi giant dyke swarm (specifically, the Nandurbar-Dhule swarm) experiencing N-S-directed crustal extension, but barring these few examples, the genetic and stratigraphic links of the Nandurbar-Dhule (and possibly Dediapada) dykes are with dykes, sills, and lava flows found in the northern and northeastern Deccan regions (such as Pachmarhi, Rewa-Shahdol, and Jabalpur-Mandla). The combined results of several studies, including the present, thus show that the stratigraphic development of the northern and northeastern Deccan is largely independent of that of the Western Ghats.

7 | CONCLUSIONS

The ENE-WSW-trending Nandurbar-Dhule dyke swarm is the best developed linear tholeiitic swarm in the Deccan Traps continental flood basalt province, with 210 dykes 1–79 km long (Ray et al., 2007). We have obtained $^{40}\text{Ar}/^{39}\text{Ar}$ ages on the swarm, of 67.06 ± 0.60 , 67.49 ± 0.89 , and 63.43 ± 0.48 Ma (2σ errors including errors on J values). These ages imply that dyke emplacement occurred over ≥ 2.5 Myr under directed lithospheric extension. Our new Sr-Nd-Pb isotopic data on these dykes, combined with previously available major and trace element (and a few isotopic) data, show that some individual dykes are multiple magma injections of different compositions. The data also suggest that some dykes were feeders to lavas of the Jawhar-Igatpuri formations of the lower Western Ghats sequence 200 km to the south and to one or two flows in the Palitana section in Saurashtra 200 km to the north-west. Other dykes broadly resemble the Bushe Formation of the middle Western Ghats sequence and broadly Bushe-like flows in the sections in the Satpura Range and the Narmada region, north-north-east of the swarm. Some dykes have geochemical signatures unlike any hitherto analysed lavas. Importantly, nine of the 17 analysed Nandurbar-Dhule dykes are isotopically like the Mahabaleshwar and Panhala formations of the upper Western Ghats sequence, covering a very narrow isotopic range, but have the distinctive chemical signatures of the high-TiO₂ Kolhapur Unit of the southernmost Western Ghats. These nine Nandurbar-Dhule dykes

possess a unique combination of isotopic and chemical characteristics, cross-combining features of different eruptive units in the Wai Subgroup of the Western Ghats. We name this geochemical flavour, not hitherto known in the Deccan tholeiites, as “Nandurbar-type.” We show that some dyke-sill intrusions in the Pachmarhi and Rewa-Shahdol areas of the Narmada-Tapi giant dyke swarm, Deccan dykes found in the Early Cretaceous Rajmahal Traps in eastern India, as well as some lava flows forming the upper parts of sections in the Mandla lobe, have strong geochemical (isotopic plus chemical) similarities to the “Nandurbar-type” dykes that dominate the Nandurbar-Dhule swarm. However, this geochemical signature of the dykes, repeatedly produced over several million years, is not found in the Western Ghats. Varied evidence accumulated now indicates that the stratigraphic development of the northern and northeastern Deccan regions (such as the Pavagadh section, Pachmarhi and Rewa-Shahdol, and the Mandla lobe) is largely independent of that of the Western Ghats.

ACKNOWLEDGEMENTS

Field work was supported by grant 03IR014 to Sheth from the Industrial Research and Consultancy Centre (IRCC), IIT Bombay. Monterova and Ivanov acknowledge the Russian Foundation for Basic Research (Grant 15-05-05130). Sr-Nd isotopic ratio measurements and $^{40}\text{Ar}/^{39}\text{Ar}$ dating were performed under the framework of the Centre for Geodynamics and Geochronology of the Institute of the Earth's Crust, Siberian Branch of the Russian Academy of Sciences, Irkutsk, Russia. Vanderkluyzen was supported by U.S. National Science Foundation grants EAR-0229825 and EAR-1250440. Critical reviews of an earlier version of this manuscript by Morgan Ganerød, Lucy McGee, John Gamble, and an anonymous reviewer helped to improve our presentation. The present manuscript was improved by constructively critical reviews from two anonymous journal referees; N. V. Chalapathi Rao is thanked for his efficient editorial handling.

CONFLICT OF INTEREST

The authors declare no conflict of interest.

ORCID

Hetu Sheth  <http://orcid.org/0000-0002-1237-0100>

Loïc Vanderkluyzen  <http://orcid.org/0000-0001-8263-1604>

REFERENCES

- Auden, J. B. (1949). Dykes in western India—A discussion of their relationships with the Deccan Traps. *Transactions of National Academy of Sciences of India*, 3, 23–157.
- Baksi, A. K. (2014). The Deccan Trap—Cretaceous-Palaeogene boundary connection: New $^{40}\text{Ar}/^{39}\text{Ar}$ ages and critical assessment of existing argon data pertinent to this hypothesis. In H. C. Sheth, & L. Vanderkluyzen (Eds.), *Flood Basalts of Asia. Journal of Asian Earth Sciences*, 84, 9–23.
- Baksi, A. K., Byerly, G. R., Chan, L.-H., & Farrar, E. (1994). Intracanyon flows in the Deccan province, India? Case history of the Rajahmundry Traps. *Geology*, 22, 605–608.
- Baragar, W. R. A., Ernst, R. E., Hulbert, L., & Peterson, T. (1996). Longitudinal petrochemical variations in the Mackenzie dyke swarm, northwestern Canadian shield. *Journal of Petrology*, 37, 317–359.

- Beane, J. E. (1988). *Flow stratigraphy, chemical variation and petrogenesis of Deccan flood basalts from the Western Ghats, India*. Ph.D. dissertation, Washington State University, USA.
- Beane, J. E., Turner, C. A., Hooper, P. R., Subbarao, K. V., & Walsh, J. N. (1986). Stratigraphy, composition and form of the Deccan basalts, Western Ghats, India. *Bulletin of Volcanology*, 48, 61–83.
- Bhattacharji, S., Chatterjee, N., Wampler, J. M., Nayak, P. N., & Deshmukh, S. S. (1996). Indian intraplate and continental margin rifting, lithospheric extension, and mantle upwelling in Deccan flood basalt volcanism near the K/T boundary: Evidence from mafic dyke swarms. *Journal of Geology*, 104, 379–398.
- Bhattacharji, S., Sharma, R., & Chatterjee, N. (2004). Two- and three-dimensional gravity modelling along western continental margin and intraplate Narmada-Tapti rifts: Its relevance to Deccan flood basalt volcanism. In H. C. Sheth, & L. Vanderkluyzen (Eds.), *Magmatism in India through Time. Proceedings of the Indian Academy of Sciences (Earth & Planetary Sciences)*, 113, 771–784.
- Bondre, N. R., Hart, W. K., & Sheth, H. C. (2006). Geology and geochemistry of the Sangamner mafic dyke swarm, western Deccan volcanic province, India: Implications for regional stratigraphy. *Journal of Geology*, 114, 155–170.
- Chalapathi Rao, N. V., Burgess, R., Lehmann, B., Mainkar, D., Pande, S. K., Hari, K. R., & Bodhankar, N. (2011). $^{40}\text{Ar}/^{39}\text{Ar}$ ages of mafic dykes from the Mesoproterozoic Chhatisgarh basin, Bastar craton, central India: Implications for the origin and spatial extent of the Deccan large igneous province. *Lithos*, 125, 994–1005.
- Chandrasekharam, D., Mahoney, J. J., Sheth, H. C., & Duncan, R. A. (1999). Elemental and Nd-Sr-Pb isotope geochemistry of flows and dykes from the Tapi rift, Deccan flood basalt province, India. *Journal of Volcanology and Geothermal Research*, 93, 111–123.
- Cox, K. G., & Hawkesworth, C. J. (1984). Relative contributions of crust and mantle to flood basalt magmatism, Mahabaleshwar area, Deccan Traps. *Philosophical Transactions of the Royal Society of London*, A310, 627–641.
- Cox, K. G., & Hawkesworth, C. J. (1985). Geochemical stratigraphy of the Deccan Traps at Mahabaleshwar, Western Ghats, India, with implications for open system magmatic processes. *Journal of Petrology*, 26, 355–377.
- Cucciniello, C., Choudhary, A. K., Zanetti, A., Sheth, H. C., Vichare, S., & Pereira, R. (2014). Mineralogy, geochemistry and petrogenesis of the Khopoli mafic intrusion, Deccan Traps, India. *Mineralogy and Petrology*, 108, 333–351.
- Cucciniello, C., Demonterova, E. I., Sheth, H., Pande, K., & Vijayan, A. (2015). $^{40}\text{Ar}/^{39}\text{Ar}$ geochronology and geochemistry of the Central Saurashtra mafic dyke swarm: Insights into magmatic evolution, magma transport, and dyke-flow relationships in the northwestern Deccan Traps. *Bulletin of Volcanology*, 77, art. 45. <https://doi.org/10.1007/s00445-015-0932-0>
- Deshmukh, S. S., & Sehgal, M. N. (1988). Mafic dyke swarms in Deccan volcanic province of Madhya Pradesh and Maharashtra. In K. V. Subbarao (Ed.), *Deccan flood basalts (Vol. 10)* (pp. 323–340). Memoirs: Geological Society of India.
- Dessai, A. G., & Viegas, A. A. A. (1995). Multi-generation mafic dyke swarm related to Deccan magmatism, south of Mumbai: Implications on the evolution of the western Indian continental margin. In T. C. Devaraju (Ed.), *Dyke swarms of Peninsular India (Vol. 33)* (pp. 435–451). Memoirs: Geological Society of India.
- Devey, C. W., & Cox, K. G. (1987). Relationships between crustal contamination and crystallization in continental flood basalt magmas, with special reference to the Deccan Traps of the Western Ghats, India. *Earth and Planetary Science Letters*, 84, 59–68.
- Devey, C. W., & Lightfoot, P. C. (1986). Volcanological and tectonic control of stratigraphy and structure in the western Deccan Traps. *Bulletin of Volcanology*, 48, 195–207.
- Ernst, R. E., & Buchan, K. L. (1997). Giant radiating dyke swarms: Their use in identifying pre-Mesozoic large igneous provinces and mantle plumes. In J. J. Mahoney, & M. F. Coffin (Eds.), *Large igneous provinces: Continental, oceanic, and planetary flood volcanism (Vol. 100)* (pp. 297–334). Geophysical Monographs: American Geophysical Union.
- Greenough, J. D., & Hodych, J. P. (1990). Evidence for lateral magma injection in the early Mesozoic dykes of eastern North America. In A. J. Parker, P. C. Ringwood, & D. H. Tucker (Eds.), *Mafic dykes and emplacement mechanisms* (pp. 35–46). Rotterdam: Balkema.
- Heri, A. R., Robyr, M., & Villa, I. M. (2014). Petrology and geochronology of 'muscovite age standard' B4M. *Geological Society London Special Publication*, 378, 69–78.
- Hooper, P. R., Widdowson, M., & Kelley, S. (2010). Tectonic setting and timing of the final Deccan flood basalt eruptions. *Geology*, 38, 839–842.
- Ivanov, A. V., Demonterova, E. I., Rasskazov, S. V., & Yasnygina, T. A. (2008). Low-Ti melts from the southeastern Siberian Traps large igneous province: Evidence for a water-rich mantle source? *Journal of Earth System Science*, 117, 1–21.
- Ivanov, A. V., He, H., Yang, L., Nikolaeva, I. V., & Paleskii, S. V. (2009). $^{40}\text{Ar}/^{39}\text{Ar}$ dating of intrusive magmatism in the Angara-Taseevskaya syncline and its implication for duration of magmatism of the Siberian Traps. *Journal of Asian Earth Sciences*, 35, 1–12.
- Ivanov, A. V., He, H., Yang, L., Ryabov, V. V., Shevko, A. Y., Paleskii, S. V., & Nikolaeva, I. V. (2013). Siberian Traps large igneous province: Evidence for two flood basalt pulses around the Permo-Triassic boundary and in the Middle Triassic, and contemporaneous granitic magmatism. *Earth-Science Reviews*, 122, 58–76.
- Jay, A. E., MacNiocaill, C., Widdowson, M., Self, S., & Turner, W. (2009). New palaeomagnetic data from the Mahabaleshwar plateau, Deccan flood basalt province, India: Implications for the volcanostratigraphic architecture of continental flood basalt provinces. *Journal of the Geological Society of London*, 166, 13–24.
- Jay, A. E., & Widdowson, M. (2008). Stratigraphy, structure and volcanology of the SE Deccan continental flood basalt province: Implications for eruptive extent and volumes. *Journal of the Geological Society of London*, 165, 177–188.
- Jicha, B. R., Singer, B. S., & Sobol, P. (2016). Re-evaluation of the ages of $^{40}\text{Ar}/^{39}\text{Ar}$ sanidine standards and supereruptions in the Western U.S. using a Noblesse multi-collector mass spectrometer. *Chemical Geology*, 431, 54–66.
- Kent, R. W., Pringle, M. S., Muller, R. D., Saunders, A. D., & Ghose, N. C. (2002). $^{40}\text{Ar}/^{39}\text{Ar}$ geochronology of the Rajmahal basalts, India, and their relationship to the Kerguelen plateau. *Journal of Petrology*, 43, 1141–1153.
- Kent, R. W., Saunders, A. D., Kempton, P. D., & Ghose, N. C. (1997). Rajmahal basalts, eastern India: Mantle sources and melt distribution at a volcanic rifted margin. In J. J. Mahoney, & M. F. Coffin (Eds.), *Large igneous provinces: Continental, oceanic, and planetary flood volcanism (Vol. 100)* (pp. 145–182). Geophysical Monographs: American Geophysical Union.
- Keshav, S., Sheth, H. C., & Chandrasekharam, D. (1998). Field geology, petrography, and orthopyroxene clusters of the Dhule-Parola dyke, Tapi valley, central Deccan basalt province. *Current Science*, 74, 252–254.
- Khadri, S. F. R., Subbarao, K. V., Hooper, P. R., & Walsh, J. N. (1988). Stratigraphy of the Thakurvadi Formation, western Deccan basalt province, India. In K. V. Subbarao (Ed.), *Deccan flood basalts (Vol. 10)* (pp. 280–304). Bangalore: Geological Society of India.
- Krishnamacharlu, T. (1971). Dykes around Dadiapada, Broach district, Gujarat. *Bulletin of Volcanology*, 35, 947–956.
- Lala, T., Choudhary, A. K., Patil, S. K., & Paul, D. K. (2011). Mafic dykes of Rewa Basin, central India: Implications on magma dispersal and petrogenesis. In R. K. Srivastava (Ed.), *Dyke swarms: Keys for geodynamic interpretation* (pp. 141–165). Berlin: Springer-Verlag.
- Lee, J.-Y., Marti, K., Severinghaus, J. P., Kawamura, K., Yoo, H.-S., Lee, J. B., & Kim, J. S. (2006). A redetermination of the isotopic abundances of atmospheric Ar. *Geochimica et Cosmochimica Acta*, 70, 4507–4512.

- Lightfoot, P. C., & Hawkesworth, C. J. (1988). Origin of Deccan Trap lavas: Evidence from combined trace element and Sr-, Nd- and Pb-isotope studies. *Earth and Planetary Science Letters*, 91, 89–104.
- Lightfoot, P. C., Hawkesworth, C. J., Devey, C. W., Rogers, N. W., & van Calsteren, P. W. C. (1990). Source and differentiation of Deccan Trap lavas: Implications of geochemical and mineral chemical variations. *Journal of Petrology*, 31, 1165–1200.
- Mahoney, J. J., Macdougall, J. D., Lugmair, G. W., Murali, A. V., Sankar Das, M., & Gopalan, K. (1982). Origin of the Deccan Trap flows at Mahabaleshwar inferred from Nd and Sr isotopic and chemical evidence. *Earth and Planetary Science Letters*, 60, 47–60.
- Mahoney, J. J., Sheth, H. C., Chandrasekharam, D., & Peng, Z. X. (2000). Geochemistry of flood basalts of the Toranmal section, northern Deccan Traps, India: Implications for regional Deccan stratigraphy. *Journal of Petrology*, 41, 1099–1140.
- Manhes, G., Allegre, C. J., & Provost, A. (1984). U-Th-Pb systematics of the eucrite "Juvinas". Precise age determination and evidence for exotic lead. *Geochimica et Cosmochimica Acta*, 48, 2247–2264.
- Melluso, L., Barbieri, M., & Beccaluva, L. (2004). Chemical evolution, petrogenesis, and regional chemical correlations of the flood basalt sequence in the central Deccan Traps, India. In H. C. Sheth, & K. Pande, K. (Eds.), *Magmatism in India through Time. Proceedings of the Indian Academy of Sciences (Earth & Planetary Sciences)*, 113, 587–603.
- Melluso, L., Beccaluva, L., Brotzu, P., Gregnanin, A., Gupta, A. K., Morbidelli, A., & Traversa, G. (1995). Constraints on the mantle sources of the Deccan Traps from the petrology and geochemistry of the basalts of Gujarat State (western India). *Journal of Petrology*, 36, 1393–1432.
- Melluso, L., Sethna, S. F., Morra, V., Khateeb, A., & Javeri, P. (1999). Petrology of the mafic dyke swarm of the Tapi River in the Nandurbar area (Deccan volcanic province). In K. V. Subbarao (Ed.), *Deccan volcanic province (Vol. 43–2)* (pp. 735–755). Memoirs: Geological Society of India.
- Mitchell, C., & Widdowson, M. (1991). A geological map of the southern Deccan Traps, India and its structural implications. *Journal of the Geological Society of London*, 148, 495–505.
- Najafi, S. J., Cox, K. G., & Sukheswala, R. N. (1981). Geology and geochemistry of the basalt flows (Deccan Traps) of the Mahad-Mahabaleshwar section, India. In K. V. Subbarao, & R. N. Sukheswala (Eds.), *Deccan volcanism (Vol. 3)* (pp. 300–315). Memoirs: Geological Society of India.
- Pande, K., Yatheesh, V., & Sheth, H. (2017). $^{40}\text{Ar}/^{39}\text{Ar}$ dating of the Mumbai tholeiites and Panvel flexure: Intense 62.5 ma onshore-offshore Deccan magmatism during India-Laxmi Ridge-Seychelles breakup. *Geophysical Journal International*, 210, 1160–1170.
- Pariso, L., Jourdan, F., Marzoli, A., Melluso, L., Sethna, S. F., & Bellieni, G. (2016). $^{40}\text{Ar}/^{39}\text{Ar}$ ages of alkaline and tholeiitic rocks from the northern Deccan Traps: Implications for magmatic processes and the K-Pg boundary. *Journal of the Geological Society of London*, 173, 679–688.
- Peate, D. W. (1997). The Paraná-Etendeka province. In J. J. Mahoney, & M. F. Coffin (Eds.), *Large igneous provinces: Continental, oceanic, and planetary flood volcanism (Vol. 100)* (pp. 217–245). Geophysical Monographs: American Geophysical Union.
- Peng, Z. X., Mahoney, J., Hooper, P., Harris, C., & Beane, J. (1994). A role for lower continental crust in flood basalt genesis? Isotopic and incompatible element study of the lower six formations of the western Deccan Traps. *Geochimica et Cosmochimica Acta*, 58, 267–288.
- Peng, Z. X., Mahoney, J. J., Hooper, P. R., Macdougall, J. D., & Krishnamurthy, P. (1998). Basalts of the northeastern Deccan Traps, India: Isotopic and elemental geochemistry and relation to southwestern Deccan stratigraphy. *Journal of Geophysical Research*, 103(B12), 29843–29865.
- Peng, Z. X., Mahoney, J. J., Hooper, P. R., & Vanderkluysen, L. (2014). Nd, Sr and Pb isotopic and chemical compositions of central Deccan Traps lavas and their relation to southwestern Deccan stratigraphy. In H. C. Sheth, & L. Vanderkluysen, L. (Eds.) *Flood Basalts of Asia. Journal of Asian Earth Sciences*, 84, 83–94.
- Phillips, D., & Matchan, E. L. (2013). Ultra-high precision $^{40}\text{Ar}/^{39}\text{Ar}$ ages for Fish Canyon tuff and Alder Creek rhyolite sanidine: New dating standards required? *Geochimica et Cosmochimica Acta*, 121, 229–239.
- Pin, C., Briot, D., Bassin, C., & Poitrasson, F. (1994). Concomitant separation of strontium and samarium-neodymium for isotopic analysis in silicate samples, based on specific extraction chromatography. *Analytica Chimica Acta*, 298, 209–217.
- Pin, C., & Santos-Zalduendi, J. F. (1997). Sequential separation of light rare-earth elements, thorium and uranium by miniaturized extraction chromatography: Application to isotopic analyses of silicate rocks. *Analytica Chimica Acta*, 339, 79–89.
- Rasskazov, S. V., Saranina, E. V., Demonerova, E. I., Maslovskaya, M. N., & Ivanov, A. V. (2002). Mantle components in late Cenozoic volcanics of the East Sayan (from Pb, Sr, and Nd isotopes). *Geologiya i Geofizika*, 42, 1065–1079.
- Ray, R. (2009). *Dykes in the Deccan Traps*. Ph.D. dissertation, Indian Institute of Technology Bombay, 146 p.
- Ray, R., Sheth, H. C., & Mallik, J. (2007). Structure and emplacement of the Nandurbar-Dhule mafic dyke swarm, Deccan Traps, and the tectonomagmatic evolution of flood basalts. *Bulletin of Volcanology*, 69, 531–537.
- Reidel, S. P., Camp, V. E., Tolan, T. L., & Martin, B. S. (2013). The Columbia River flood basalt province: Stratigraphy, areal extent, volume, and physical volcanology. In S. P. Reidel, V. E. Camp, M. E. Ross, J. A. Wolff, B. S. Martin, T. L. Tolan, & R. E. Wells (Eds.), *The Columbia River flood basalt province (Vol. 497)* (pp. 1–43). Special Papers: Geological Society of America.
- Renne, P. R., Balco, G., Ludwig, K. R., Mundil, R., & Min, K. (2011). Response to the comment by W. H. Schwarz et al. on "joint determination of ^{40}K decay constant and $^{40}\text{Ar}^*/^{40}\text{K}$ for the Fish Canyon sanidine standard, and improved accuracy for $^{40}\text{Ar}/^{39}\text{Ar}$ geochronology" by P.R. Renne et al. (2010). *Geochimica et Cosmochimica Acta*, 75, 5097–5100.
- Renne, P. R., Cassata, W. S., & Morgan, L. E. (2009). The isotopic composition of atmospheric argon and $^{40}\text{Ar}/^{39}\text{Ar}$ geochronology: Time for a change? *Quaternary Geochronology*, 4, 288–298.
- Renne, P. R., Mundil, R., Balco, G., Min, K., & Ludwig, K. R. (2010). Joint determination of ^{40}K decay constants and $^{40}\text{Ar}^*/^{40}\text{K}$ for the Fish Canyon sanidine standard, and improved accuracy for $^{40}\text{Ar}/^{39}\text{Ar}$ geochronology. *Geochimica et Cosmochimica Acta*, 74, 5349–5367.
- Renne, P. R., Sprain, C. J., Richards, M. A., Self, S., Vanderkluysen, L., & Pande, K. (2015). State shift in Deccan volcanism at the Cretaceous-Paleogene boundary, possibly induced by impact. *Science*, 350, 76–78.
- Richards, M. A., Alvarez, A., Self, S., Karlstrom, L., Renne, P. R., Manga, M., ... Gibson, S. A. (2015). Triggering of the largest Deccan eruptions by the Chicxulub impact. *Geological Society of America Bulletin*, 127, 1507–1520.
- Sant, D. A., & Karanth, R. V. (1990). Emplacement of dyke swarms in the Lower Narmada valley, western India. In A. J. Parker, P. C. Rickwood, & D. H. Tucker (Eds.), *Mafic dykes and emplacement mechanisms* (pp. 383–389). Rotterdam: Balkema.
- Schoene, B., Samperton, K. M., Eddy, M. P., Keller, G., Adatte, T., Bowring, S. A., ... Gertsch, B. (2015). U-Pb geochronology of the Deccan Traps and relation to the end-Cretaceous mass extinction. *Science*, 347, 182–184.
- Self, S., Jay, A. E., Widdowson, M., & Keszthelyi, L. P. (2008). Correlation of the Deccan and Rajahmundry Trap lavas: Are these the longest and largest lava flows on Earth? *Journal of Volcanology and Geothermal Research*, 172, 3–19.
- Sen, G., & Cohen, T. H. (1994). Deccan intrusion, crustal extension, doming and the size of the Deccan-Réunion plume head. In K. V. Subbarao (Ed.), *Volcanism* (pp. 201–216). New Delhi: Wiley Eastern.
- Sheth, H. C., Duncan, R. A., Chandrasekharam, D., & Mahoney, J. J. (1997). Deccan Trap dioritic gabbros from the western Satpura-Tapi region. *Current Science*, 72, 755–757.
- Sheth, H. C., Mahoney, J. J., & Chandrasekharam, D. (2004). Geochemical stratigraphy of flood basalts of the Bijasan Ghat section, Satpura Range, India. *Journal of Asian Earth Sciences*, 23, 127–139.

- Sheth, H. C., & Melluso, L. (2008). The Mount Pavagadh volcanic suite, Deccan Traps: Geochemical stratigraphy and magmatic evolution. *Journal of Asian Earth Sciences*, 32, 5–21.
- Sheth, H. C., Ray, J. S., Ray, R., Vanderkluyzen, L., Mahoney, J. J., Kumar, A., ... Jana, B. (2009). Geology and geochemistry of Pachmarhi dykes and sills, Satpura Gondwana Basin, central India: Problems of dyke-sill-flow correlations in the Deccan Traps. *Contributions to Mineralogy and Petrology*, 158, 357–380.
- Sheth, H. C., Zellmer, G. F., Demonterova, E. I., Ivanov, A. V., Kumar, R., & Patel, R. K. (2014). The Deccan tholeiite lavas and dykes of Ghatkopar-Powai area, Mumbai, Panvel flexure zone: Geochemistry, stratigraphic status, and tectonic significance. In: H. C. Sheth, & L. Vanderkluyzen, L. (Eds.), *Flood Basalts of Asia*. *Journal of Asian Earth Sciences*, 84, 69–82.
- Sheth, H. C., Zellmer, G. F., Kshirsagar, P. V., & Cucciniello, C. (2013). Geochemistry of the Palitana flood basalt sequence and the Eastern Saurashtra dykes, Deccan Traps: Clues to petrogenesis, dyke–flow relationships, and regional lava stratigraphy. *Bulletin of Volcanology*, 75, art. 701. <https://doi.org/10.1007/s00445-013-0701-x>
- Shrivastava, J. P., Mahoney, J. J., & Kashyap, M. R. (2014). Trace elemental and Nd-Sr-Pb isotopic compositional variation in 37 lava flows of the Mandla lobe and their chemical relation to the western Deccan stratigraphic succession, India. *Mineralogy and Petrology*, 108, 801–817.
- Steiger, R. H., & Jager, E. (1977). Subcommittee on geochronology: Convention on the use of decay constants in geo- and cosmochemistry. *Earth and Planetary Science Letters*, 36, 359–362.
- Subbarao, K. V., Chandrasekharam, D., Navaneethakrishnan, P., & Hooper, P. R. (1994). Stratigraphy and structure of parts of the central Deccan basalt province: Eruptive models. In K. V. Subbarao (Ed.), *Volcanism* (pp. 321–332). New Delhi: Wiley Eastern.
- Subbarao, K. V., & Hooper, P. R. (1988). Reconnaissance map of the Deccan basalt group in the Western Ghats, India. In K. V. Subbarao (Ed.), *Deccan flood basalts (Vol. 10) (enclosure)*. Memoirs: Geological Society of India.
- Subbarao, K. V., Hooper, P. R., Dayal, A. M., Walsh, J. N., & Gopalan, K. (1999). Narmada dykes. In K. V. Subbarao (Ed.), *Deccan volcanic province (Vol. 43–2)* (pp. 891–902). Memoirs: Geological Society of India.
- Sun, S.-S., & McDonough, W. F. (1989). Chemical and isotopic systematics of oceanic basalts: Implications for mantle composition and processes. In A. D. Saunders, & M. J. Norry (Eds.), *Magmatism in the ocean basins (Vol. 42)* (pp. 313–345). Geological Society: Special Publications.
- Swanson, D. A., Wright, R. L., & Helz, R. T. (1975). Linear vent systems and estimated rates of magma production and eruption for the Yakima Basalt on the Columbia Plateau. *American Journal of Science*, 275, 877–905.
- Vanderkluyzen, L., Mahoney, J. J., Hooper, P. R., Sheth, H. C., & Ray, R. (2011). The feeder system of the Deccan Traps (India): Insights from dyke geochemistry. *Journal of Petrology*, 52, 315–343.
- Vernikovsky, V. A., Pearse, V. L., Vernikovskaya, A. E., Romanov, A. P., Gee, D. G., & Travin, A. V. (2003). First report of early Triassic A-type granite and syenite intrusions from Taimyr: Product of the northern Eurasian superplume? *Lithos*, 66, 23–36.
- Widdowson, M., Pringle, M. S., & Fernandez, O. A. (2000). A post-K-T boundary (Early Palaeocene) age for Deccan-type feeder dykes, Goa, India. *Journal of Petrology*, 41, 1177–1194.

SUPPORTING INFORMATION

Additional Supporting Information may be found online in the supporting information tab for this article.

How to cite this article: Sheth H, Vanderkluyzen L, Demonterova EI, Ivanov AV, Savatenkov VM. Geochemistry and $^{40}\text{Ar}/^{39}\text{Ar}$ geochronology of the Nandurbar-Dhule mafic dyke swarm: Dyke-sill-flow correlations and stratigraphic development across the Deccan flood basalt province. *Geological Journal*. 2019;54:157–176. <https://doi.org/10.1002/gj.3167>